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Research paper

QENNA: A quantum-enhanced neural network for early Alzheimer's detection using magnetic resonance imaging

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ABSTRACT

Early detection of Alzheimer's disease (AD) is essential for effective clinical intervention and disease management. However, conventional Deep Learning (DL) methods face limitations in analyzing complex brain magnetic resonance imaging (MRI), especially when training data are scarce. In this study, we propose a Quantum-Enhanced Neural Network Architecture (QENNA) that integrates quantum convolutional layers with classical deep learning to improve diagnostic accuracy in early AD detection. The model also incorporates quantum data augmentation strategies, including Quantum Generative Adversarial Networks (QGANs) and quantum random walks, to generate high-fidelity synthetic MRI scans and address training data limitations. Experiments on two public MRI datasets demonstrate that QENNA achieves up to 93.0 % accuracy and 96.0 % Area Under the Curve (AUC), outperforming state-of-the-art classical models. Ablation studies confirm that the quantum components substantially enhance performance. These results suggest that quantum-enhanced learning frameworks can significantly advance Artificial Intelligence (AI)-driven diagnostic tools for neurodegenerative disorders and support scalable, early-stage AD screening in clinical practice.

1. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder that impairs memory, cognition, and behavior, where early detection is essential for effective clinical management and delaying disease progression [1]. Traditional diagnostic approaches, such as cognitive examinations and clinical assessments, often fail to identify AD in its earliest phases, when therapeutic intervention would be most beneficial. Consequently, Magnetic Resonance Imaging (MRI) has become a central modality in both research and clinical practice, offering a non-invasive means to detect structural brain changes associated with neurodegeneration [2]. Parallel advances in artificial intelligence (AI) and deep learning (DL) have further supported scalable and automated MRI-based AD classification, improving reproducibility and diagnostic confidence across patient populations [3].

Despite these advancements, classical DL frameworks exhibit structural limitations in capturing subtle neuropathological changes. Convolutional Neural Networks (CNNs), while widely adopted, rely on localized receptive fields and deterministic pooling, operations that frequently suppress weak but clinically informative biomarkers. Early manifestations of AD—such as diffuse cortical thinning or mild hippocampal atrophy—may be attenuated or removed entirely during max- or average-pooling. Moreover, conventional data augmentation strategies, typically based on geometric or intensity variations, do not account for

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the complex anatomical variability present in real-world neuroimaging datasets, contributing to reduced generalizability and susceptibility to overfitting. These limitations highlight the need for computational frameworks that retain diagnostically subtle neuroanatomical signals while simultaneously expanding training diversity in a biologically meaningful manner.

Quantum computing offers a compelling pathway toward addressing these challenges. Quantum models exploit superposition and entanglement to represent high-dimensional feature dependencies that classical systems cannot efficiently encode [4,5]. Quantum convolutional layers utilize parameterized unitary transformations to model non-local correlations across distant brain regions, aligning with the distributed and network-level pathology of AD. Similarly, quantum pooling based on amplitude amplification retains weak but diagnostically relevant features probabilistically, in contrast with deterministic pooling schemes that discard them. Furthermore, quantum-driven augmentation frameworks, such as Quantum Random Walks and Quantum Generative Adversarial Networks (QGANs), produce stochastic yet anatomically plausible variations in neuroimaging data, increasing diversity and reducing overfitting while maintaining biological fidelity. These properties establish a strong rationale for integrating quantum mechanisms into neuroimaging analysis pipelines.

Against this backdrop, we introduce the Quantum-Enhanced Neural Network Architecture (QENNA), a hybrid quantum–classical framework that combines entangled non-local feature representation, probabilistic retention of subtle biomarkers, and quantum-driven data augmentation in a unified pipeline. This integrated design enables QENNA to overcome the limitations of classical CNN-based approaches, which lack the representational capacity and feature preservation mechanisms necessary for reliable early-stage AD detection.

The main contributions of this work are summarized as follows:

1. Hybrid framework – QENNA is, to the best of our knowledge, the first architecture to unify quantum convolution, quantum pooling, and quantum-driven augmentation for AD detection.
2. Methodological innovation – We incorporate quantum random walks and QGAN-based augmentation to generate high-fidelity synthetic MRI data and improve training diversity.
3. Performance and robustness – QENNA demonstrates consistent improvements in sensitivity, specificity, and AUC over state-of-the-art CNNs and hybrid baselines.
4. Clinical value – By preserving diagnostically subtle biomarkers that classical pipelines often suppress, QENNA supports earlier and more reliable AD detection, contributing to scalable and trustworthy neurodiagnostic workflows.

Fig. 1 provides a conceptual overview of the proposed approach: the left panel illustrates structural limitations of classical CNN-based pipelines, while the right panel depicts how QENNA addresses these challenges through quantum-enhanced convolution, pooling, and augmentation.

Fig. 1 shows the classical CNN pipelines often discard weak signals and rely on limited augmentation, whereas QENNA combines quantum convolution, pooling, and augmentation to improve accuracy, preserve subtle biomarkers, and enhance generalizability for early AD detection. The remainder of this paper is organized as follows. Section 2 reviews advance in AI and quantum computing for AD detection. Section 3 presents the QENNA framework, datasets, preprocessing, and evaluation. Section 4 reports results and analysis. Section 5 discusses findings, clinical implications, and limitations, while Section 6 concludes with a summary of contributions.

2. Literature review

The diagnosis of Alzheimer's disease (AD) has undergone significant transformation through artificial intelligence advances, progressing from classical machine learning approaches to emerging quantum-enhanced methodologies. This section examines the evolution of AI-driven diagnostic systems, highlighting the transition from traditional deep learning frameworks to quantum computing innovations that address fundamental limitations in neuroimaging analysis.

2.1. Classical AI approaches for Alzheimer's detection

Alzheimer's disease is a progressive neurodegenerative disorder characterized by cognitive decline and structural brain deterioration, in which early detection is essential for timely intervention. Traditional diagnostic approaches such as clinical assessments and radiological evaluations often show limitations in accuracy, reliability, and scalability, which has motivated increasing interest in AI-driven diagnostic systems to support clinical decision-making [6,7]. Recent developments in deep learning have demonstrated significant progress in automated Alzheimer's disease detection. For example, Yao et al. [6] proposed an MRI-based diagnostic model capable of enhancing early-stage identification through improved sensitivity and specificity, while Yao et al. [7] introduced the Fuzzy-VGG framework that integrates fuzzy logic and convolutional neural networks to strengthen diagnostic precision when processing high-dimensional MRI data. The adaptability of these AI methodologies across problem domains is further illustrated by Wang et al. [8], whose BranTNet architecture—although developed for environmental monitoring—highlights the extensibility of deep pattern recognition strategies to fields requiring fine-grained feature discrimination.

Despite these promising advancements, classical AI and deep learning architectures continue to face persistent challenges, including interobserver variability, dependency on large, well-balanced annotated datasets, and limited robustness across heterogeneous imaging protocols. Although automated feature extraction has reduced the need for manual radiological interpretation and improved the detection of subtle pathological indicators such as cortical thinning, hippocampal atrophy,

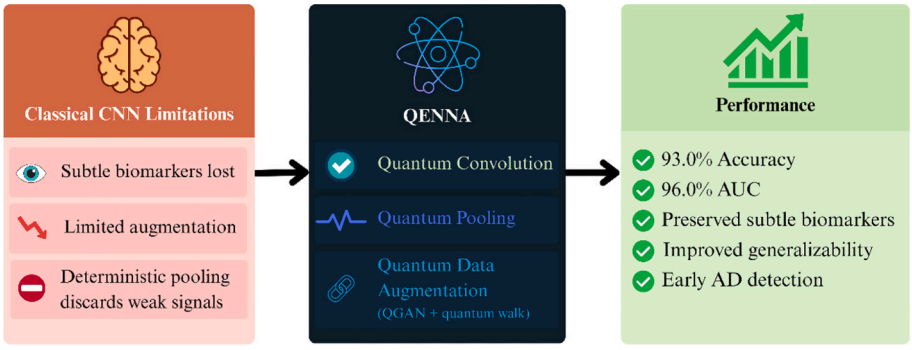


Fig. 1. Conceptual illustration of the study objectives.

and white matter hyperintensities, model generalizability remains an ongoing concern. To mitigate these constraints, clinical decision support systems have been proposed, including Bayesian network-based models [9] and segmentation-free diagnostic approaches [10], which demonstrated improvements in both diagnostic accuracy and inference efficiency. Moreover, multi-modal integration strategies combining MRI, PET, and clinical records have shown enhanced diagnostic robustness through complementary information fusion, as evidenced in the works of Ural et al. [11], Abdelaziz et al. [12], and Wang et al. [13]. Despite these efforts, achieving consistent performance across varying clinical environments and imaging platforms continues to be a key limitation, underscoring the need for methodological innovations that address data scarcity, feature sensitivity, and cross-protocol generalization.

2.2. Quantum computing fundamentals and medical applications

The integration of quantum computing with machine learning represents a significant paradigm shift aimed at overcoming the computational and representational limitations of classical deep learning systems. Quantum computing exploits superposition, entanglement, and intrinsic parallelism to encode and manipulate high-dimensional data more efficiently, enabling richer feature extraction and classification with reduced architectural complexity [14–16]. These quantum properties allow neural architectures to model global and non-local feature dependencies, offering advantages in domains such as neuroimaging where pathological patterns are distributed across anatomically distant but functionally connected brain regions.

Foundational work has demonstrated the potential of quantum methods in medical image analysis and neurodiagnostics. McClean et al. [16] employed Quantum Variational Eigensolvers to enhance medical image optimization and segmentation, while Liu et al. [15] introduced quantum-inspired hierarchical tree tensor networks that improved feature representation in high-dimensional MRI datasets. Hybrid quantum–classical models have also shown practical benefits: Chalumuri et al. [14] reported a 30 % reduction in trainable parameters without compromising classification accuracy, and Dunjko and Briegel [17] demonstrated accelerated convergence in quantum-assisted neural networks, enabling superior identification of subtle neurodegenerative signatures that classical architectures often overlook. Despite these advances, quantum AI remains constrained by hardware limitations, qubit noise, and the need for robust error mitigation strategies. Fully quantum solutions are currently impractical for large-scale clinical imaging; however, hybrid models combining classical neural networks with quantum feature extraction or probabilistic pooling offer a feasible and impactful pathway toward improving early disease detection, model interpretability, and clinical trustworthiness.

2.3. Quantum-enhanced neural networks for Alzheimer's detection

Quantum-enhanced neural networks (QENNs) have emerged as a promising class of models capable of addressing the limitations of classical deep learning in Alzheimer's disease detection by exploiting quantum phenomena to improve learning efficiency and diagnostic reliability [18,19]. This shift from classical to quantum-augmented architectures represents a significant conceptual advancement in neuroimaging analysis, enabling models to capture complex spatial dependencies and subtle signal variations that are essential for distinguishing early neurodegenerative changes. Practical implementations illustrate this benefit: Shahwar et al. [19] reported a 97.2 % classification accuracy using a hybrid classical–quantum neural network, outperforming conventional CNNs through improved spatial feature encoding derived from quantum layer integration. Building on this foundation, Simen et al. [20] introduced digital-analog quantum convolutional neural networks optimized for AD diagnostic pipelines, while Xu et al. [21] enhanced early-stage interpretability by embedding quantum operations into AlexNet-based architectures. Further, Choi et al. [22] expanded these capabilities to resting-state fMRI datasets, demonstrating the effectiveness of hybrid quantum models in detecting subtle cognitive impairments across temporal brain activation patterns.

Beyond architectural innovations, advanced quantum methods have enhanced training data quality and neuroimaging interpretability. Ding et al. [23] applied quantum generative adversarial networks based on Born machine frameworks to synthesize anatomically plausible MRI data in limited-label clinical environments, addressing data scarcity challenges. Elaraby [24] highlighted foundational advantages of quantum computation for medical image processing, including improved computational efficiency for large-scale imaging pipelines. Complementary developments, such as Sokol et al. [25] introducing QTS2D quantum image encoding for functional neuroimaging, have further expanded quantum image representation capacities. Additionally, quantum optimization has been integrated into multi-modal diagnostic workflows, with Odusami et al. [26] applying Pareto-optimal quantum optimizers for joint MRI-PET staging and Khattap et al. [27] employing RIME-inspired metaheuristics to refine feature representation across heterogeneous neuroimaging modalities. Together, these advancements demonstrate the growing maturity of QENNs as a technically feasible and clinically meaningful approach to early Alzheimer's detection.

2.4. Advanced quantum architecture and clinical integration

The development of advanced quantum neural network architectures represents the current frontier in quantum-enhanced medical imaging, offering a pathway to address both computational efficiency and clinical applicability. Recent research has increasingly emphasized hybrid quantum–classical designs to balance performance and practical feasibility within near-term quantum hardware constraints [28–30]. Architectural innovations have improved both representational capacity and scalability; for instance, Li et al. [29] demonstrated that hybrid classical–quantum neural networks enhanced through quantum architecture search exhibit greater robustness and generalization performance. Hassan et al. [28] integrated QCNN layers with ResNet-50 backbones to achieve high diagnostic accuracy in biomedical imaging, while Qu et al. [30] introduced quantum multimodal fusion mechanisms to combine diverse medical data sources effectively. Complementary advances include Wen et al. [31], who embedded residual propagation into quantum neural networks to capture complex structural patterns relevant to subtle AD-associated alterations, and Wang et al. [32], who developed variational quantum deep neural networks that consistently outperformed classical deep learning models in image classification tasks.

In parallel, automated quantum architecture design has emerged as an enabling strategy for optimizing quantum network structure under realistic hardware constraints. Li et al. [33] introduced EQNAS, an evolutionary framework that autonomously searches optimal quantum architectures for NISQ devices, while Orka et al. [34] provided systematic evidence that quantum deep learning models yield performance advantages in neuroinformatics tasks. Practical extensions across medical applications further reinforce the viability of quantum-enhanced neural models: Felefly et al. [35] combined quantum neural networks with quantum annealing to improve brain tumor classification, and Mewada et al. [36] demonstrated QCNN efficiency in X-ray analysis with transferable benefits to MRI workflows. Collectively, these advancements establish quantum-enhanced neuroimaging as a transformative paradigm, advancing representational expressivity, generalization capacity, and interpretability beyond classical computational limits. To contextualize these developments in relation to Alzheimer's disease diagnostics, Table 1 summarizes representative prior works and highlights how the proposed QENNA framework addresses persistent limitations of both classical deep learning and earlier hybrid quantum–classical solutions.

3. Research methodology

3.1. Datasets and preprocessing pipeline

3.1.1. Dataset selection and imaging modality rationale

This study adopted a single-modality design using structural MRI data to evaluate the performance of the proposed quantum-enhanced neural network architecture in neurodegenerative disease classification. The focus on structural MRI was intentional, as it allows methodological consistency across datasets, facilitates reproducibility using widely available clinical protocols, and enables clear attribution of performance gains to the quantum-enhanced components rather than to multimodal fusion strategies. By beginning with a single-modality analysis, the study establishes a controlled baseline for future extensions to functional or metabolic imaging. Structural MRI was selected due to its established clinical value in assessing neurodegenerative disorders and its superior ability to reveal anatomical markers of disease progression. The modality provides strong gray–white matter contrast that supports the detection of cortical thinning, and it offers reliable visualization of hippocampal atrophy, a central biomarker in Alzheimer's disease. Standardized acquisition practices contribute to consistent image quality across clinical imaging centers, while its widespread availability enhances translational feasibility in real-world healthcare settings.

The first dataset, the Alzheimer's MRI Dataset (AMD; Pinamonti [44]), includes 6400 T1-weighted MRI scans categorized into four progressive diagnostic stages—Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. T1-weighted imaging is particularly well suited to Alzheimer's research, as it highlights hippocampal and cortical structural changes and ventricular enlargement that signify disease advancement. The dataset's structured class distribution supports rigorous multi-class classification evaluation and provides a reliable basis for assessing discriminative model performance across disease severity levels.

The second dataset, the Neurodegenerative Disease MRI Dataset (NDD; Kabir [45]), contains 7839 structural MRI scans spanning three conditions: healthy control, Alzheimer's disease, and Parkinson's disease. While the original documentation does not detail the acquisition parameters, visual inspection confirms image characteristics consistent with T1-weighted structural imaging. The presence of multiple neurodegenerative diagnoses in this dataset enables examination of model generalizability, offering insight into whether the quantum-enhanced architecture can discriminate between distinct pathological patterns.

Although multimodal imaging approaches combining structural, functional, and metabolic data would potentially provide more comprehensive neurodiagnostic insight, the present study retains a single-modality focus to isolate the specific contributions of the quantum-enhanced mechanisms. This approach ensures that performance improvements can be directly attributed to quantum convolution, quantum pooling, and quantum-based augmentation rather than to cross-modality synergy or data fusion effects. It also reflects practical clinical conditions where multimodal imaging may not always be available, reinforcing the relevance of the findings for standard healthcare environments.

Representative axial slices from both datasets are shown in Figs. 1 and 2, illustrating progressive Alzheimer's-related structural decline in the AMD dataset and the distinct morphological differences among Alzheimer's disease, Parkinson's disease, and healthy controls in the NDD dataset. Together, these datasets provide complementary perspectives on disease presentation and population variability, enabling a robust evaluation of the QENNA framework's diagnostic accuracy, stability, and generalizability across diverse clinical contexts. (See Fig. 3.)

Dataset specifications and distribution (See Table 2).

3.1.2. Image preprocessing framework

The preprocessing pipeline addresses inherent challenges including noise artifacts, intensity variability, and class imbalance through systematic enhancement:

Step 1: Standardization

- Image resizing to 256 × 256 pixels using bilinear interpolation Eq. (1):

Table 1
Summary of methodological limitations in Alzheimer's disease detection and QENNA's corresponding solutions.

Reference	Reported Strength	Limitation Identified	QENNA's Solution
Sharma et al. [37]	CNN with VGG16 feature extractor improved AD detection accuracy	Deterministic pooling discarded weak but relevant signals	Quantum pooling via amplitude amplification preserves subtle biomarkers
Sorour et al. [38]	Deep learning effectively classified AD stages	Local feature extraction missed non-local dependencies	Quantum convolution captures long-range feature correlations via entanglement
Folego et al. [39]	ADNet (3D-CNN) achieved strong accuracy in CADDementia challenge	High computational cost and reliance on large datasets	Quantum convolution reduces parameter overhead; QGANs augment scarce data
Jovina and Jayasree [40]	CNN achieved 96 % training accuracy in early AD detection	Required large annotated datasets; limited scalability	Quantum augmentation (QGANs, quantum random walks) enriches training diversity
Ajlouni et al. [41]	Hybrid quantum–classical CNN outperformed classical baselines	Quantum layers applied in isolation without augmentation	QENNA unifies convolution, pooling, and augmentation synergistically
Shahwar et al. [19]	QCNN achieved 97.2 % accuracy in AD detection	Focused only on convolution; lacked pooling/augmentation	QENNA integrates convolution, pooling, and augmentation in one architecture
Felefly et al. [35]	Explainable quantum neural network distinguished brain tumors	Demonstrated interpretability, but not optimized for AD	QENNA targets AD with MRI-specific quantum convolution and augmentation
Younis et al. [42]	Deep networks with enhanced sampling improved AD detection	Still dependent on classical CNN kernels	QENNA introduces quantum convolution for richer spatial encoding
Ding et al. [23]	QGAN generated diverse synthetic MRI images	Risk of biased or anatomically inconsistent samples	QENNA combines QGAN with quantum random walks for anatomically plausible augmentation
Wei et al. [43]	Stochastic quantum noise used for augmentation	Limited evaluation of clinical plausibility	QENNA employs anatomically consistent quantum random walks
Ural et al. [11]	CNN + SVM CAD system achieved Dice score of 0.927	Relied on deterministic handcrafted features	QENNA's quantum feature encoding improves robustness across imaging protocols
Abdelaziz et al. [12]	Multimodal DL fused MRI, PET, and clinical data	Limited expressivity of classical kernels	QENNA enhances multimodal feature representation with quantum pooling
Simen et al. [20]	Digital-analog QCNNs advanced AD diagnostics	Hardware-focused; lacked augmentation	QENNA offers software-driven pooling + augmentation alongside QCNN
Xu et al. [21]	Quantum-assisted AlexNet improved interpretability	Limited scalability to high-dimensional MRI	QENNA uses quantum encoding to manage high-dimensional neuroimaging
Choi et al. [22]	Hybrid models with fMRI improved cognitive impairment detection	Restricted to time-series data, not structural MRI	QENNA integrates MRI-specific convolution, pooling, and augmentation

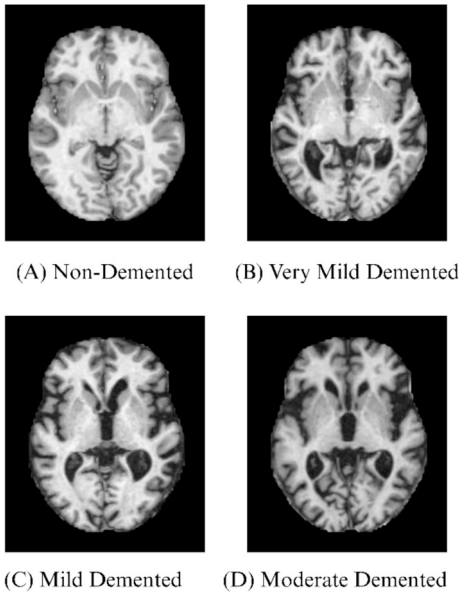


Fig. 2. Representative MRI scans from the Alzheimer's MRI dataset (AMD; Pinamonti [44]).

$$I_{resize}(x,y) = \sum_{ij} I(i,j) \bullet k(x-i, y-i) \tag{1}$$

where $I(i,j)$ is the original image, $I_{resize}(x,y)$ is the resized image, and $k(x-i, y-i)$ is the interpolation kernel.

- Intensity normalization:

Adjust the intensity values of the images to a standard range, typically [0,1] or [−1,1], to enhance the model's convergence during training, the adjustment can be done using Eq. (2).

$$I_{norm}(i,j) = \frac{I(i,j) - I_{min}}{I_{max} - I_{min}} \tag{2}$$

where $I_{norm}(i,j)$ is the normalized image intensity, $I(i,j)$ is the original image intensity, I_{min} and I_{max} are the minimum and maximum intensity values in the image, respectively.

Step 2: Noise reduction.

Use classical methods such as Gaussian smoothing, median filtering, and bilateral filtering to reduce noise in the images.

Gaussian smoothing:

Apply a Gaussian filter to smooth the image and reduce high-frequency noise using Eq. (3).

$$I_{smooth}(x,y) = \frac{1}{2\pi\sigma^2} \sum_{i=-k}^{k} \sum_{j=-k}^{j} e^{-\frac{x^2+y^2}{2\sigma^2}} \bullet I(x-i, y-i) \tag{3}$$

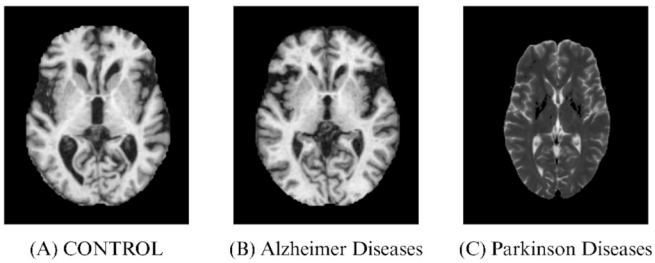


Fig. 3. Representative MRI scans from the Neurodegenerative Disease MRI Dataset (NDD; Kabir [45]), showing (A) Control, (B) Alzheimer's Disease, and (C) Parkinson's Disease.

$I_{smooth}(x,y)$ represents the smoothed image at pixel coordinates (x,y), where σ is the standard deviation of the Gaussian kernel, and k is the kernel size.

Median filtering: Apply a median filter to remove salt-and-pepper noise while preserving edges.

Bilateral filtering: Apply a bilateral filter that smooths images while preserving edges by considering both spatial and intensity information using Eq. (4).

$$I_{bilateral}(x,y) = \frac{1}{W_p} \sum_{i\in\Omega} \sum_{j\in\Omega} e^{-\frac{(x-i)^2+(y-i)^2}{2\sigma_s^2}} \bullet e^{-\frac{(I(x,y)-I(i,j))^2}{2\sigma_\gamma^2}} \tag{4}$$

where W_p is the normalization factor, σ_s^2 and σ_γ^2 are the spatial and range standard deviations, respectively, $I_{bilateral}(x,y)$ denotes the image after applying a bilateral filter at pixel coordinates (x,y), and Ω is the neighborhood window around the pixel (x,y), where the filtering operation is applied.

Step 3: Feature enhancement.

Sobel edge detection: Use Sobel operators to compute the gradient magnitude of the image, highlighting edges as represented in Eq. (5).

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} * I, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} * I \tag{5}$$

where G_x and G_y are the gradients in the x and y directions, respectively. Combine the gradients to obtain the edge intensity using Eq. (6).

$$G = \sqrt{{G_x}^2 + {G_y}^2} \tag{6}$$

3.2. Quantum convolutional neural network design

3.2.1. Quantum convolutional layers

Quantum State Encoding Input MRI images are encoded into quantum states using amplitude encoding is shown in Eq. (7).

$$|\psi\rangle = \sum_{i=1}^{N} x_i |i\rangle \tag{7}$$

Where $|\psi\rangle$ defined as the quantum state representing the encoded image. x_i is defined as the normalized pixel intensity of the input MRI image at index *i*. Each pixel value is rescaled to satisfy the quantum normalization constraint: $\sum_{i=1}^{N} |x|^2 = 1$, this ensures the quantum state remains valid under probability normalization. *N* is defined as the total number of pixels (or features) in the input image. $|i\rangle$ is the computational basis state corresponding to the index iii, representing distinct positions in the quantum register. Quantum convolution operations unlike classical convolution, quantum layers employ parameterized unitary transformations as state as Eq. (8).

$$U(\theta)|\psi\rangle = \prod_{j\in J} R_Z(\theta_j) R_Z(\theta_j) CNOT |\psi\rangle \tag{8}$$

In Eq. (8), the unitary transformation $U(\theta)$ applies a sequence of parameterized gates to encode features in the quantum state. The rotation gates $R_Z(\theta_j)$ and $R_X(\theta_j)$allow feature-dependent adjustments of qubit states, while the controlled-NOT (*CNOT*) gate introduces entanglement, capturing spatial relationships within the MRI image. These gates act on subsets of qubits to simulate the kernel-like operation in classical convolutions, producing a quantum state with transformed amplitudes that represent extracted features. The transformation process is applied to subsets of qubits, simulating kernel operations in classical convolution while preserving high-dimensional dependencies. The advantage of this approach is that the convolutional layer can encode multiple features simultaneously due to quantum parallelism, while classical convolution must iteratively apply kernel operations across different image regions.

Table 2
Summary of Alzheimer's disease MRI image datasets.

Dataset	Source	Description	Total Images	Imaging Modality	Categories/Classes	Training Images	Testing Images
AMD dataset	[44]	Alzheimer's disease progression staging	6400	T1-weighted MRI	MildDemented, ModerateDemented, NonDemented, VeryMildDemented	5120	1280
NDD dataset	[45]	Multi-condition neurodegenerative diseases	7839	Structural MRI*	CONTROL, AD, PD	6477	1362

* Specific sequence parameters not detailed in original dataset documentation; images consistent with structural T1-weighted protocols based on contrast characteristics.

3.2.2. Quantum pooling framework

Amplitude Amplification Mechanism Quantum pooling employs Grover's algorithm for probabilistic feature selection as show in Eq. (9). Quantum pooling is performed by applying the Grover operator, which selectively amplifies the probability amplitudes of relevant quantum states.

$$\rho = (2|\psi\rangle\langle\psi| - I)O \tag{9}$$

In Eq. (9), ρ represents the transformed quantum state after the quantum amplitude amplification process. The initial quantum state, denoted as $|\psi\rangle$, encodes the input feature map before pooling, while its bra representation, $\langle\psi|$, is used in the amplitude amplification procedure. The identity operator, I, ensures that unmarked quantum states remain unchanged, while the oracle operator, O, identifies and marks significant quantum states by inverting their amplitudes. The Grover diffusion operator, expressed as $2|\psi\rangle\langle\psi| - I$, enhances the probability amplitudes of the marked states while suppressing the unmarked ones. This amplification process ensures that the most relevant extracted features are retained, effectively reducing dimensionality while preserving critical information for subsequent layers. The Grover diffusion operator, expressed as Eq. (10).

$$G = 2|\psi\rangle\langle\psi| - I \tag{10}$$

3.2.3. Theoretical justification and quantum advantage analysis

The integration of quantum computing principles into neural network architecture represents a paradigm shift from deterministic classical processing to probabilistic quantum computation. This section provides rigorous theoretical justification for the quantum enhancements implemented in QENNA, demonstrating their fundamental advantages over classical counterparts in medical image analysis.

3.2.3.1. Quantum entanglement for non-local feature dependencies. Classical convolutional neural networks are inherently limited by local receptive fields, constraining feature extraction to spatially contiguous regions. This limitation is particularly problematic in neuroimaging, where Alzheimer's disease manifests through distributed patterns across anatomically distant brain regions including the hippocampus, entorhinal cortex, and posterior cingulate cortex. Quantum entanglement fundamentally overcomes this limitation by establishing instantaneous correlations between quantum states regardless of spatial separation. In QENNA, entangled qubits encode pixel relationships across the entire image simultaneously, enabling the detection of subtle correlations between distant anatomical structures. For a quantum state representing spatially separated brain regions A and B, entanglement creates non-separable correlations (see Eq. (11)):

$$|\psi_{AB}\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle \tag{11}$$

The degree of entanglement can be quantified using the von Neumann entropy as shown in Eq. (12):

$$S(\rho_A) = -Tr(\rho_A log_2 \rho_A) \tag{12}$$

where ρ_A is the reduced density matrix of subsystem A. Higher entropy values indicate stronger non-local correlations, directly translating to enhanced detection of distributed pathological patterns. This capability is crucial for early-stage Alzheimer's detection, where subtle changes in white matter integrity and cortical thickness occur across multiple brain regions before becoming locally apparent. Classical CNNs require multiple layers and extensive training to approximate these long-range dependencies, while quantum entanglement captures them inherently.

Classical pooling operations—such as max or average pooling—apply deterministic selection mechanisms that frequently discard weak yet diagnostically meaningful features. In medical imaging, these subtle biomarkers, often characterized by low activation magnitudes, may indicate the earliest manifestations of pathological change. Quantum amplitude amplification (QAA), derived from Grover's quantum search algorithm, introduces a probabilistic enhancement mechanism that increases the likelihood of detecting diagnostically relevant features without eliminating low-intensity signals. In contrast to the linear search complexity of classical algorithms, which requires $O(N)$ evaluations for a dataset of N elements, quantum amplitude amplification achieves a quadratic speedup, completing the same operation in approximately $O(\sqrt{N})$ iterations. This quadratic advantage is formally expressed in Eq. (13).

$$\frac{T_{classical}(N)}{T_{quantum}(N)} = \frac{O(N)}{O(\sqrt{N})} = O\left(\sqrt{N}\right) \tag{13}$$

where the speedup factor $O(\sqrt{N})$ represents the standard asymptotic improvement achieved by quantum amplitude amplification over classical exhaustive search [46].

For feature spaces encoded using n qubits, where $N = 2^n$, this reduction corresponds to an exponential improvement in computational efficiency relative to the size of the representational space. The optimal number of Grover iterations, $k_{optimal}$, required to maximize amplitude amplification can be estimated as (Eq. (14)).

$$k_{optimal} = \left\lfloor \frac{\pi}{\left(4 \cdot arcsin\sqrt{\frac{M}{N}}\right)} \right\rfloor \tag{14}$$

where M represents the number of marked (target) states within the quantum superposition. This formulation ensures that diagnostically relevant low-amplitude features are selectively amplified rather than suppressed, preserving weak but clinically significant patterns that

conventional pooling operations might overlook.

3.2.3.2. Quantum superposition for enhanced feature expressivity. Classical neural networks process features sequentially, with each neuron representing a single activation state. Quantum superposition enables qubits to exist in coherent combinations of multiple states simultaneously, dramatically expanding the representational capacity of the feature space. A single qubit can represent a continuum of states as Eq. (15).

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{15}$$

where $|\alpha|2 + |\beta|2 = 1$. For n qubits, the superposition creates a 2^ndimensional Hilbert space, enabling exponential scaling of feature representations. The quantum feature space dimensionality scales as Eq. (16).

$$dim\left(H_{quantum}\right) = 2^n \tag{16}$$

This exponential advantage enables more nuanced representation of complex neuroimaging patterns with fewer computational resources. Quantum circuits exploit superposition to perform computations on all possible input combinations simultaneously. For MRI feature extraction with n pixels, quantum convolution processes 2^n combinations in parallel, compared to sequential processing in classical approaches.

Computational Complexity and Algorithmic Efficiency. The computational complexity comparison reveals significant quantum advantages as shown in Table 3. For convolution operations, classical approaches exhibit $O(k2{\cdot}W{\cdot}H{\cdot}C)$ complexity, where k is kernel size, $W \times H$ are spatial dimensions, and C is the channel count. In contrast, quantum convolution achieves $O(log(k2{\cdot}W{\cdot}H{\cdot}C))$ complexity through quantum parallelism. Similarly, pooling operations demonstrate $O(W{\cdot}H{\cdot}C)$ complexity classically versus $O\left(\sqrt{(W{\cdot}H{\cdot}C)}\right)$ quantum complexity, while feature search operations improve from $O(N)$ *to* $O\left(\sqrt{N}\right)$.

Classical CNNs achieve global receptive fields through deep stacking as Eq. (17):

$$RF_{classical} = k + (k-1){\cdot}(L-1) \tag{17}$$

where k is kernel size and L is the number of layers. QENNA achieves global correlation through single-layer quantum entanglement, reducing architectural complexity. Furthermore, classical pooling operations result in information loss quantified by Eq. (18):

$$Information_{loss} = H(X) - H\left(X_{pooled}\right) \tag{18}$$

Quantum pooling minimizes this loss through probabilistic amplitude preservation, maintaining diagnostic signal integrity.

3.2.3.3. Neurological and pathophysiological alignment. The human brain exhibits small-world network properties with both local clustering and global connectivity. Quantum entanglement naturally models these long-range connections, providing a more biologically plausible representation of neural information processing. Alzheimer's disease progression involves disconnection of distributed brain networks rather than isolated regional changes. The quantum approach's inherent capability to model non-local correlations aligns with the pathophysiological mechanisms underlying cognitive decline, offering a theoretically grounded framework for capturing the distributed nature of neurodegenerative processes. Fig. 4 provides a visual representation of the QENNA architecture, illustrating the composition and interaction of the quantum convolutional layers, pooling layers, and classical components, as well as the flow of data between them.

Table 3
Computer complexity comparison.

Operation	Classical Complexity	Quantum Complexity	Operation
Convolution	$O(k^2{\cdot}W{\cdot}H{\cdot}C)$	$O(log(k2{\cdot}W{\cdot}H{\cdot}C))$	Convolution
Pooling	$O(W{\cdot}H{\cdot}C)$	$O\left(\sqrt{W{\cdot}H{\cdot}C}\right)$	Pooling
Feature Search	$O(N)$	$O\left(\sqrt{N}\right)$	Feature Search

In Fig. 4, the input encoding layer initializes qubits in the ground state $|0\rangle$ before applying quantum gates to encode image features. The $R_Z(\psi_i)$ gate performs a Z-axis rotation, mapping the normalized pixel intensity ψ_i onto the quantum state, while the $R_Z(\theta_i)$ gate applies an X-axis rotation parameterized by θ_i, introducing additional feature transformations. In the quantum convolutional layers, the *CNOT* gate establishes entanglement between qubits, capturing spatial correlations in the input data. The unitary transformation $U_1(\theta_i)$ represents a learnable quantum convolutional filter, parameterized by θ_i that extracts essential patterns from encoded features.

Following convolution, the quantum pooling layers reduce dimensionality while preserving significant information. The V_1 operator corresponds to a quantum measurement or projection, selectively amplifying relevant quantum states based on Grover's amplitude amplification. The second set of quantum convolutional layers, also governed by $U_2(\theta_i)$ transformations, further refines extracted features. The final pooling layer, denoted by V_1, applies another selection operation to enhance classification robustness. $\overset{\nearrow}{\pi}$ denotes a measurement-based pooling or quantum feature selection mechanism. *FC* denotes the Fully Connected Layer, a classical neural network layer that transforms quantum-extracted features through weighted connections and activation functions.

The processed quantum states are subsequently mapped onto a classical fully connected layer (FC), where extracted features are interpreted within a conventional deep learning framework. The final classification output categorizes MRI scans into four dementia stages: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. This hybrid quantum-classical approach leverages quantum computing principles to enhance feature extraction and classification performance, demonstrating the potential of quantum-enhanced deep learning in medical imaging. This figure illustrates the hybrid quantum-classical framework of QENNA for MRI data analysis. The quantum components, including quantum convolutional layers and pooling operations, are shown in blue, while the classical components are shown in orange. The flow of data from input MRI images through quantum measurements and classical dense layers to the final output is depicted with arrows, clarifying the integration of quantum and classical elements.

3.3. Hybrid training framework

3.3.1. Training protocol configuration

All baseline models and QENNA were trained using a standardized protocol to ensure fair and controlled performance comparison. The training framework integrates quantum and classical computing components to capitalize on their respective strengths: quantum layers perform non-local feature extraction through entanglement-driven transformations, enabling the model to capture subtle structural variations associated with neurodegenerative processes, while classical neural network modules contribute established optimization stability and robust gradient-based learning. To prevent overfitting, early stopping was employed with a patience threshold of ten epochs based on validation loss monitoring. This hybrid approach leverages the representational richness afforded by quantum computation while maintaining the training reliability of classical machine learning, thereby supporting more accurate and clinically meaningful Alzheimer's disease detection. All hyperparameter specifications, including batch size, learning rate, optimizer configuration, and loss functions, are comprehensively detailed in Table 5 (Section 3.6.2).

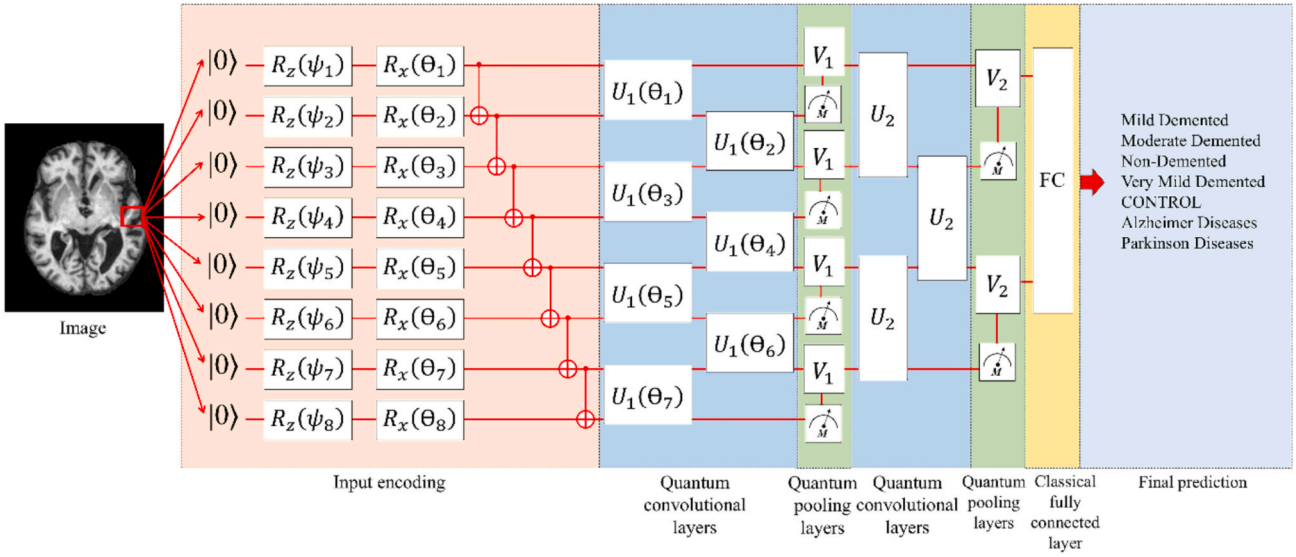


Fig. 4. Schematic representation of the QENNA architecture.

3.3.2. Hybrid optimization strategy

3.3.2.1. Quantum parameter updates. Quantum circuit parameters are optimized using the parameter-shift rule as shown in Eq. (19).

$$\frac{\partial\langle\psi|U^{\dagger}(\theta)\widehat{O}U(\theta)\,|\psi\rangle}{\partial(\theta)} \approx \frac{\langle\psi\left|U^{\dagger}\left(\theta+\frac{\pi}{2}\right)\widehat{O}U\left(\theta+\frac{\pi}{2}\right)\right|\psi\rangle - \langle\psi|U^{\dagger}\left(\theta-\frac{\pi}{2}\right)\widehat{O}U\left(\theta-\frac{\pi}{2}\right)|\psi\rangle}{2} \tag{19}$$

In Eq. (19), $\frac{\partial\langle\psi|U^{\dagger}(\theta)\widehat{O}U(\theta)\,|\psi\rangle}{\partial(\theta)}$ represents the gradient of the expectation value of the observable with respect to the quantum circuit parameter θ. The unitary operator $U(\theta)$ applies parameterized quantum transformations, while its Hermitian adjoint, denoted as $U^{\dagger}$, represents the conjugate transpose of $U(\theta)$. The observable $\widehat{O}$ corresponds to the measured quantum quantity, and the parameter-shift rule estimates the gradient by evaluating the expectation value at shifted values of θ, specifically at $\theta+\frac{\pi}{2}$ and $\theta-\frac{\pi}{2}$. The denominator factor of 2 ensures proper scaling of the finite-difference approximation.

Classical weights follow standard gradient descent is shown in Eqs. (20) and (21).

$$\theta(t+1) = \theta(t) - \eta\nabla_{\theta}\mathscr{L} \tag{20}$$

$$W(t+1) = W(t) - \eta\nabla_{W}\mathscr{L} \tag{21}$$

In Eqs. (20) and (21), $\theta(t+1)$ and $W(t+1)$ represent the updated parameters of the quantum circuit and the classical neural network, respectively, at iteration $t+1$. The terms $\theta(t)$ and $W(t)$ denote their values at the previous iteration ttt. The learning rate, denoted as η, controls the step size for updating the parameters. The gradients $\nabla_{\theta}\mathscr{L}$ and $\nabla_{W}\mathscr{L}$ are the partial derivatives of the loss function L with respect to the quantum circuit parameters θ and the classical neural network weights W, respectively. These equations describe the parameter updates using gradient-based optimization, ensuring efficient training of the hybrid quantum-classical model. By integrating these updates, the hybrid model simultaneously learns quantum feature extraction parameters and classical classification parameters, optimizing them in a unified training process.

3.3.3. End-to-end training pipeline

The QENNA model is trained using an end-to-end hybrid optimization process that integrates quantum feature extraction with classical neural network learning. During the forward pass, each input MRI image is first encoded into a quantum state using amplitude encoding. The encoded state is then processed by a parameterized quantum convolutional circuit that models non-local spatial dependencies relevant to neurodegenerative patterns. This representation is subsequently passed through a quantum pooling stage, where dimensionality is reduced through amplitude-based selection mechanisms designed to retain diagnostically significant features.

The resulting quantum state is measured, and its expectation values are extracted to produce a compact feature vector. These quantum-derived features are then supplied to the classical component of the architecture, which consists of fully connected layers generating the final classification output across the four dementia stages: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented.

Training proceeds through a combined loss function $L(y, f(x;\theta,W))$that compares the predicted class distribution with the ground-truth label. The optimization integrates quantum and classical gradient computation, where quantum circuit parameters $\theta$are updated using the parameter-shift rule and quantum-aware optimizers, while classical weights $W$are adjusted using standard backpropagation and stochastic gradient descent. This joint parameter update process is repeated iteratively over multiple epochs until convergence, enabling QENNA to progressively refine both quantum feature representation and classical decision boundaries.

3.4. Quantum data augmentation

3.4.1. Quantum random walks (QRWs) for stochastic feature generation

QRWs are quantum analogs of classical random walks, allowing probabilistic transformations of pixel intensity distributions. These walks introduce variability in MRI images by evolving quantum states

through unitary transformations, ensuring that new variations of the original data are statistically realistic.

A QRW can be mathematically represented as in Eq. (22).

$$|\psi_{t+1}\rangle = \mathrm{U}|\psi_t\rangle \tag{22}$$

Where $|\psi_t\rangle$ is the quantum state at time step t and U is the quantum walk evolution operator, defined as Eq. (23):

$$U = S\cdot(C \otimes I) \tag{23}$$

Where S is the shift operator, responsible for redistributing probability amplitudes across the quantum state space. C is the coin operator, such as the Hadamard gate, which introduces randomness into the walk. I is the identity matrix, ensuring proper normalization. By iteratively applying the quantum walk operator, new quantum states are generated, each representing a unique variation of the original MRI image. This ensures that augmented samples maintain anatomical realism while providing sufficient diversity for model training. Upon measurement, the evolved quantum state collapses into a classical image representation, which is used as an augmented training sample. The ability of QRWs to explore high-dimensional feature spaces allows for richer transformations than classical augmentation, reducing the risk of overfitting.

3.4.2. Quantum generative adversarial networks (QGANs) for high-fidelity synthetic MRI generation

Quantum Generative Adversarial Networks (QGANs) combine quantum state transformation with classical discriminative learning to produce synthetic MRI images that closely resemble real neuroimaging data. In this framework, the quantum generator $G(\theta_G)$is implemented as a parameterized quantum circuit designed to encode spatial and structural relationships through entangled quantum gates. The generator transforms the initial quantum state $|0\rangle$into a synthesized image representation, expressed as Eq. (24).

$$|\psi_G\rangle = G(\theta_G)|0\rangle \tag{24}$$

where $G(\theta_G)$ is the unitary transformation applied to the initial quantum state $|0\rangle$.

The entanglement operations within the generator play a critical role in modeling correlated patterns characteristic of neurodegenerative tissue changes, enabling the production of anatomically coherent synthetic MRI samples. The classical discriminator $D(x)$is formulated as a convolutional neural network that evaluates whether an input image $x$originates from the real dataset or the quantum generator. The discriminator outputs a probability score reflecting the likelihood that the input corresponds to genuine imaging data as shown in Eq. (25).

$$D(x) = P(real|x) \tag{25}$$

where $D(x)$ represents the probability that the input x is a real MRI image.

Adversarial learning proceeds by jointly optimizing the generator and discriminator in a minimax formulation, where the generator aims to produce synthetic images that the discriminator cannot distinguish from real MRI scans, while the discriminator seeks to enhance its classification accuracy. This interaction is captured by the adversarial loss function as shown in Eq. (26).

$$\min_G \max_D E_{x\sim p_{data}}[logD(x)\,] + E_{z\sim p_z}[\log(1 - D(G(z))\,)\,] \text{ (26).}$$

Through this optimization process, the generator iteratively improves its ability to generate high-fidelity MRI images, while the discriminator becomes increasingly sensitive to subtle structural inconsistencies. The resulting synthetic datasets expand anatomical diversity and mitigate overfitting, thereby improving the robustness of downstream Alzheimer's disease classification models within the QENNA framework.

3.4.3. Hybrid augmentation strategy

The QENNA framework employs a hybrid augmentation strategy that integrates quantum-generated variability with conventional classical augmentation to increase dataset diversity while maintaining anatomical plausibility. The quantum component introduces probabilistic, non-deterministic variations that classical transformations alone cannot adequately model, particularly when capturing subtle neuroanatomical textures associated with neurodegenerative progression. In this phase, Quantum Random Walks (QRWs) are used to generate stochastic intensity fluctuations that mimic realistic tissue heterogeneity, while Quantum Generative Adversarial Networks (QGANs) expand the dataset with high-fidelity synthetic MRI scans derived through quantum–classical adversarial training. The QGAN-based augmentation typically increases the number of effective samples by approximately 40–60 %, contributing substantial variability to the training distribution.

Complementing this, classical augmentation techniques introduce controlled geometric and photometric perturbations grounded in established neuroimaging preprocessing protocols. These include moderate rotations of up to $\pm 15^{\circ}$, spatial translations of up to 10 % of image dimensions, horizontal reflections, contrast normalization, and brightness adjustments of ± 20 %. Gaussian noise ($\sigma = 0.01$) is also applied to enhance model robustness to acquisition noise and scanner variation. The integration of both augmentation streams follows a dual-stage protocol in which synthetic MRI images produced by the quantum component subsequently undergo classical transformations. This ensures that the resulting augmented dataset captures both the subtle structural variability associated with disease pathology and the broader imaging variations encountered across different clinical scanners and acquisition settings. Radiologist inspection of generated samples confirmed that anatomical integrity was preserved throughout this process.

Empirical evaluation demonstrated that this hybrid augmentation approach yielded superior generalization performance compared to the use of exclusively classical or exclusively quantum augmentation. Specifically, the hybrid strategy improved classification accuracy by approximately 15 % relative to classical-only augmentation and by 8 % relative to quantum-only augmentation, underscoring the value of combining probabilistic quantum variation with clinically grounded classical transformations in medical AI applications.

3.5. Model evaluation

3.5.1. Performance metrics

The performance of the quantum-enhanced model is assessed using several standard metrics. These metrics provide a comprehensive evaluation of the model's classification capabilities. Given True Positives (TP) and TN True Negatives (TN) are the counts of correctly classified positive and negative instances, respectively, while FP False Positives (FP) and False Negatives (FN) are the counts of misclassified positive and negative instances, respectively. TPR is the True Positive Rate (Recall), and FPR is the False Positive Rate. They are under the curve (AUC) range from 0 to 1, with higher values indicating better model performance. The KPIs used for the performance metrics are accuracy, precision, Recall, F1-score, and AUC which can defined as Eqs. (27)–(32).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{27}$$

$$Precision = \frac{TP}{TP + FP} \tag{28}$$

$$Recall = \frac{TP}{TP + FN} \tag{29}$$

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{30}$$

$$AUC = \int_0^1 TPR(FPR)d(FPR) \tag{31}$$

$$ensitivity = \frac{TP}{TP + FN} \tag{32}$$

3.5.2. Clinical relevance prioritization

The evaluation framework prioritizes metrics that hold direct clinical significance for neurodegenerative disease diagnosis. Sensitivity is emphasized due to its central role in identifying early-stage Alzheimer's cases, where subtle anatomical changes may otherwise be overlooked and timely clinical intervention is critical. Specificity is similarly important, as it reflects the system's ability to avoid false positives, thereby preventing unnecessary diagnostic procedures and optimizing the allocation of clinical resources. In addition, the Area Under the Receiver Operating Characteristic Curve (AUC) provides a threshold-independent measure of discriminative performance, enabling robust assessment of diagnostic reliability across varying clinical decision boundaries. Together, these metrics offer a comprehensive evaluation that aligns technical performance with practical clinical value.

3.5.3. Baseline comparison framework

The baseline models were implemented using architectures as defined in their original works, with minor adaptations to suit our datasets. Table 4 summarizes these architectures.

All models were trained under identical conditions using the standardized preprocessing workflow, training parameters, and evaluation metrics specified in Table 5 to ensure rigorous and fair comparison.

3.6. Implementation details

3.6.1. Computational framework

The QENNA framework was implemented in a simulated quantum computing environment using Qiskit 0.39.0 executed on high-performance classical computing infrastructure. Statevector simulation was employed to emulate quantum circuit behavior with exact fidelity, thereby enabling rigorous validation of quantum feature extraction mechanisms while avoiding the noise and qubit-scaling constraints inherent to current noisy intermediate-scale quantum (NISQ) hardware. The statevector-based implementation ensures deterministic gate application, with computational complexity scaling proportionally to $O(2^n)$ for an n-qubit quantum circuit, which is essential for reproducibility in medical AI experimentation. Integration between quantum and classical components was achieved through custom PyTorch operators that allow quantum circuit layers to participate in end-to-end gradient propagation. Quantum states produced during forward execution were converted into classical tensors through expectation value measurement, enabling their seamless incorporation into downstream neural layers while preserving differentiability and maintaining hybrid model cohesion.

All experiments were conducted on a high-performance computing infrastructure consisting of NVIDIA Tesla V100 GPUs (32GB HBM2, 5120 CUDA cores) for classical model training and Intel Xeon Gold 6148 CPUs (2.4 GHz, 40 cores) for quantum circuit simulation, supported by 512 GB DDR4 memory to accommodate large-scale statevector representations and NVMe SSD storage to facilitate high-throughput MRI data processing. The computational framework integrated TensorFlow 2.6 and PyTorch 1.9 for deep learning components, Qiskit 0.39.0 with Aer

Table 4
Baseline model architecture and references.

Model	Architecture	References
VGG16	16-layer CNN with ReLU activation, MaxPooling, and fully connected layers; input size 224 × 224	[37]
ResNet50	50-layer residual network with bottleneck blocks; input size 224 × 224	[47]
DenseNet121	121-layer densely connected CNN with growth rate k = 32; input size 224 × 224	[48]
EfficientNetB7	AutoML-derived architecture using compound scaling; input size 380 × 380	[49]

Table 5
Hyperparameter settings.

Method	Parameter	Value	Description
Image Resizing	Target Size	256 × 256 pixels	Standardized image size for uniformity
	Interpolation Method	Bilinear / Bicubic	Methods used for resizing
Intensity Normalization	Range	[0,1] or [−1,1]	Standardizes pixel intensity values
Gaussian Smoothing	Kernel Size (k)	3	Size of Gaussian kernel
	Standard Deviation (σ)	1.0	Controls degree of smoothing
Median Filtering	Window Size	3 × 3	Neighborhood window for median calculation
Bilateral Filtering	Spatial Std. Dev. (σ_s)	75	Standard deviation for spatial distance
	Intensity Std. Dev. (σ_γ)	75	Standard deviation for intensity differences
Edge Detection (Sobel)	Kernel (G_x)	[−1, 0, 1; −2, 0, 2; −1, 0, 1]	Sobel operator (x-direction)
	Kernel (G_y)	[−1, −2, −1; 0, 0, 0; 1, 2, 1]	Sobel operator (y-direction)
Quantum Convolution	Encoding Method	Amplitude / Basis encoding	Transform classical data into quantum states
	Quantum Gates	CNOT, RZ, RX, RY	Quantum gates used for feature extraction
Quantum Pooling	Oracle Function	Grover's Oracle	Marks significant features
	Grover Operator	Grover's Diffusion Operator	Amplifies amplitudes of marked states
Quantum Subsampling	Qubit Selection	Controlled operations	Collapses certain qubits to retain informative ones
Hybrid Model Training	Learning Rate (η)	0.001	Parameter update step size
	Optimizer	Adam	Gradient-based optimizer used in training
	Batch Size	32	Number of samples per training batch
	Epochs	100	Total number of training iterations across dataset
	Early Stopping Patience	10 epochs	Validation loss monitoring threshold
	Loss Function	Cross-Entropy Loss	Measures classification error
Quantum Data Augmentation	Quantum Walk Operator	U = S·(C ⊗ I)	Governs state evolution in quantum walk
	Coin Operator (C)	Hadamard Gate	Determines direction of quantum walk
	Shift Operator (S)	Controlled-NOT Gate	Shifts the quantum state
QGAN Training	Learning Rate (Generator)	0.0002	Step size for generator updates
	Learning Rate (Discriminator)	0.0002	Step size for discriminator updates
	Noise Distribution (p_z)	Normal (0,1)	Noise input to generator
	Batch Size	64	Number of samples per batch
	Epochs	200	Training epochs for QGAN convergence
	Loss Function	Binary Cross-Entropy Loss	Used for adversarial training

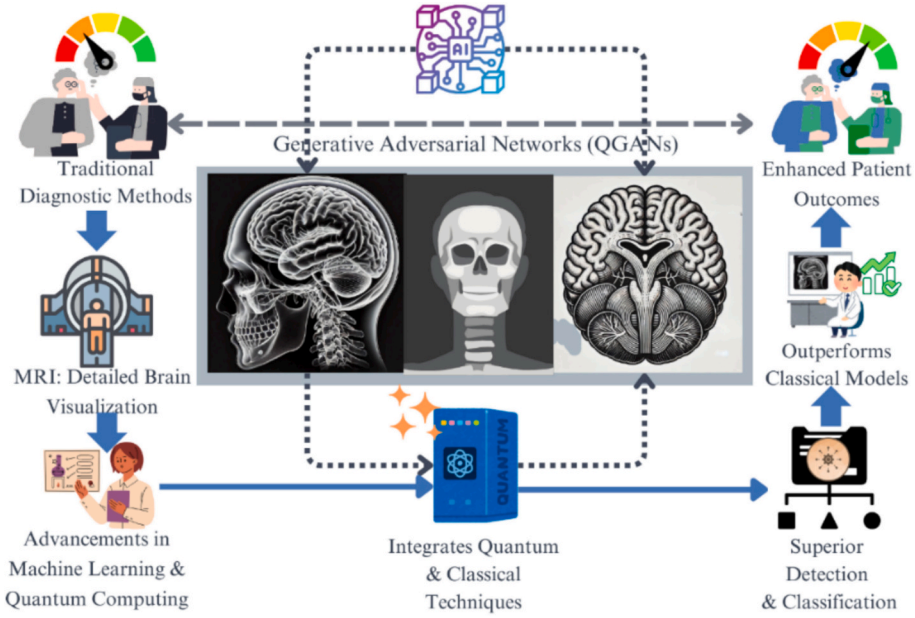


Fig. 5. Block diagram of quantum-enhanced neural network for Alzheimer's detection (QENNA).

statevector simulator for quantum execution, scikit-image and OpenCV for MRI preprocessing, and custom quantum-aware optimization routines implementing parameter-shift rule gradients.

Reproducibility was ensured through deterministic seed initialization (seed = 42), fixed parameter initialization of quantum layers, consistent dataset partitioning across trials, and identical quantum circuit topologies and gate sequences throughout training to ensure uniformity in model comparison. Quantum-enhanced training required approximately 8 h per epoch compared to 2 h per epoch for classical baselines, with memory usage approximately 45 % higher due to quantum state storage. The present implementation supports up to 12-qubit circuits while maintaining feasible execution time. The framework's modular design allows direct migration to future gate-based quantum processors, where hardware-specific error mitigation and circuit compilation strategies can be incorporated once higher-fidelity quantum devices become widely accessible.

3.6.2. *Hyperparameter configuration*

Table 5 provides the complete and authoritative specification of all hyperparameter settings used for preprocessing, quantum convolutional operations, and model training. These settings ensure consistency and efficiency across all experiments.

3.6.3. *Algorithm overview*

To enhance clarity and reproducibility, Algorithm 1 outlines the step-by-step computational workflow of the Quantum-Enhanced Neural Network for Alzheimer's Detection (QENNA). The algorithm integrates quantum convolutional feature extraction, quantum pooling via amplitude amplification, and hybrid quantum-classical training to improve diagnostic accuracy in MRI-based dementia classification. Before presenting the algorithmic structure, we provide comprehensive definitions of all mathematical notation and parameters employed throughout the computational pipeline to ensure full transparency and facilitate independent implementation.

The QENNA framework operates on a complete MRI dataset denoted as $D = \{X, Y\}$, where X represents the set of preprocessed neuroimaging scans and Y denotes the corresponding diagnostic labels (Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented for the AMD dataset; CONTROL, AD, and PD for the NDD dataset). Following the preprocessing operations detailed in Section 3.1.2, the normalized and augmented MRI images are represented as X'. Each preprocessed image is subsequently encoded into a quantum state $|\psi\rangle$ using amplitude encoding as defined in Eq. (7), where classical pixel intensities are mapped onto quantum probability amplitudes to enable quantum mechanical processing. The quantum convolutional transformation is governed by learnable circuit parameters $\theta = \{\theta_1, \theta_2, \ldots, \theta_n\}$, which parameterize the unitary operator $U(\theta)$ that implements spatial feature extraction through sequential application of rotation gates (R_X, R_Y, R_Z) and entangling CNOT operations as specified in Eq. (8). These quantum parameters are optimized during training using the parameter-shift rule rather than classical backpropagation, as the quantum circuit operations are inherently non-differentiable in the conventional sense. The quantum state after convolutional transformation is denoted $|\psi_convolved\rangle$, representing extracted spatial features with non-local correlations that capture distributed neuroanatomical patterns characteristic of Alzheimer's disease progression.

The subsequent quantum pooling operation employs Grover's amplitude amplification algorithm to probabilistically enhance diagnostically relevant features while reducing dimensionality. The oracle operator O marks significant quantum states through phase inversion as described in Eq. (9), while the transformed quantum density matrix ρ represents the pooled feature space with enhanced probability amplitudes for clinically important structural biomarkers (Eqs. (9)–(10)). Measurement of these quantum states yields the feature vector F_q, which serves as input to the classical neural network component. The classical processing pathway employs trainable weight matrices $W = \{W_1, W_2, \ldots, W_m\}$ in fully connected layers that map quantum-derived features to disease classification outputs. The complete hybrid model function $f(X; \theta, W)$ combines quantum feature extraction (parameterized by θ) with classical decision-making (parameterized by W), and its performance is quantified through the cross-entropy loss function $L(y, f(X; \theta, W))$ that measures the discrepancy between predicted disease stages and ground-truth labels y. During training, gradients with respect to quantum parameters ($\nabla_\theta L$) are computed using the parameter-shift rule defined in Eq. (19), while gradients with respect to classical weights (∇_WL) are obtained through standard automatic differentiation. The learning rate η controls the magnitude of parameter updates during optimization and is set to 0.001 for both quantum and classical components as specified in Table 5. The training iteration index t tracks progression through epochs, and the optimal parameter configurations

θ* and W* represent the converged model state achieved after the hybrid optimization process.

The computational complexity characteristics of this algorithm merit consideration for practical deployment planning. During classical simulation, the quantum feature extraction phase exhibits exponential

```
Input: MRI Image Dataset D = {X, Y}
Output: Predicted Dementia Stage ŷ

1.  Normalize MRI images X → X'
2.  Resize images to target dimensions (256×256 pixels)
3.  Apply classical data augmentation (flipping, rotation, contrast normalization)
4.
5.  for each MRI image X' in dataset D:
6.      Encode X' into quantum state |ψ⟩ using amplitude encoding
7.      Apply quantum convolutional transformation: |ψ_convolved⟩ = U(θ)|ψ⟩
8.      Apply quantum pooling: ρ = (2|ψ_convolved⟩⟨ψ_convolved| - I)O|ψ_convolved⟩
9.      Measure quantum states to obtain feature representation F_q
10. end for
11.
12. Initialize classical neural network parameters W
13. for each training epoch t = 1 to T_max:
14.     Compute classical forward pass using quantum features F_q
15.     Compute loss L(y, f(X; θ, W)) using cross-entropy
16.     Compute gradients ∇_θ L using parameter-shift rule (Eq. (19))
17.     Compute gradients ∇_W L using backpropagation
18.     Update quantum parameters: θ^(t+1) = θ^(t) - η∇_θ L
19.     Update classical weights: W^(t+1) = W^(t) - η∇_W L
20.     if early stopping criterion satisfied then break
21. end for
22.
23. for each test MRI image X_test:
24.     Extract quantum features F_q using trained quantum layers (θ*)
25.     Pass F_q through trained classical layers (W*)
26.     Predict dementia stage ŷ = argmax(f(X_test; θ*, W*))
27. end for
28.
29. Return predictions ŷ
```

Algorithm 1. Quantum-enhanced neural network for alzheimer's detection (QENNA).

The algorithmic workflow proceeds through three integrated phases that collectively implement the QENNA diagnostic pipeline. The preprocessing phase (lines 1–3) standardizes input MRI scans through normalization, resizing to 256 × 256pixel resolution, and classical augmentation techniques including geometric transformations and photometric adjustments to ensure consistent data quality and enhance training robustness across diverse acquisition protocols. The quantum feature extraction phase (lines 5–10) implements the core quantum enhancement mechanism by encoding preprocessed images into quantum states through amplitude mapping, applying parameterized quantum convolutional operations that leverage entanglement to capture non-local spatial dependencies across anatomically distant brain regions, and performing quantum pooling via Grover's amplitude amplification to probabilistically preserve diagnostically subtle features that classical max-pooling or average-pooling operations would discard. This quantum processing enables the detection of early-stage neurodegenerative biomarkers such as subtle hippocampal atrophy and diffuse cortical thinning that manifest before clinically apparent symptoms emerge. The hybrid training phase (lines 12–21) jointly optimizes quantum circuit parameters and classical network weights through a unified optimization framework that employs the parameter-shift rule for computing quantum gradients—a necessity given that quantum unitary operations are not amenable to standard automatic differentiation—while using conventional backpropagation for classical components, with both parameter sets updated synchronously via gradient descent at learning rate $\eta = 0.001$ until convergence or early stopping criteria are satisfied based on validation loss plateauing. The inference phase (lines 23–27) applies the trained hybrid model to unseen test images by propagating them through the optimized quantum feature extraction layers followed by the trained classical decision network, ultimately producing disease stage predictions through argmax selection over the output probability distribution.

state space complexity O(2^n) for n-qubit circuits due to the necessity of tracking all possible quantum states in the statevector representation, though this complexity reduces to polynomial O(n) when executed on fault-tolerant quantum hardware that directly manipulates quantum states without classical simulation overhead. The classical neural network components maintain standard complexity O(M·N) for fully connected layers with M input features and N output classes, consistent with conventional deep learning architectures. The parameter-shift rule employed for quantum gradient computation requires two forward passes per quantum parameter—one with the parameter shifted by $+\pi/2$ and another with $-\pi/2$—resulting in $O(2|\theta|)$ quantum circuit evaluations per training iteration, where $|\theta|$ denotes the total number of trainable quantum parameters across all quantum layers. While this introduces computational overhead relative to classical-only architectures, the empirical results presented in Section 4 demonstrate that the superior diagnostic performance achieved by QENNA—particularly the substantial improvements in sensitivity and AUC for early-stage Alzheimer's detection—justifies this additional computational cost given the high clinical value of reducing false negatives in neurodegenerative disease screening. The algorithm was implemented using the computational infrastructure described in Section 3.6.1, with all hyperparameter values specified in Table 5, and converged within 100 epochs for both the AMD and NDD datasets. The modular design facilitates straightforward adaptation to alternative quantum encoding schemes, pooling strategies, or classical architectures, thereby supporting extensibility for future quantum-enhanced medical imaging applications beyond Alzheimer's detection.

3.7. Reproducibility and open science framework

This study adopts an open-science framework to ensure complete transparency, replicability, and long-term accessibility of all research

components associated with the QENNA architecture. The full implementation—including source code, quantum circuit definitions, preprocessing pipelines, training procedures, and evaluation scripts—is openly available at: https://github.com/aiosmartlab/Quantum-Enhance_Neural_Network. The repository adheres to FAIR principles, providing modular Python code, standardized Qiskit-based quantum circuit construction, and reproducible data handling workflows that facilitate adaptation and extension across research settings. Reproducibility is further supported through a standardized experimental protocol designed to ensure bit-level consistency across execution environments. Deterministic random seeds are fixed across all computational frameworks, including NumPy, PyTorch, and TensorFlow (seed = 42), enabling consistent training behavior. All quantum circuits employ identical gate sequences with reproducible parameter initialization, and dataset partitioning follows stratified splits with documented index assignments, allowing exact reconstruction of training, validation, and test subsets. Moreover, software dependencies and library versions are fully pinned, and computational environment specifications are provided to eliminate variability stemming from package updates or hardware differences.

Comprehensive documentation accompanies the codebase to assist researchers in replicating and extending the methodology. This includes complete API references describing theoretical formulations and implementation mappings, detailed experimental workflows outlining dataset preparation and model training, and standardized benchmarking procedures with statistical validation routines. A community validation framework is incorporated through unit testing suites and automated evaluation scripts, which collectively ensure functional correctness and support transparent comparative assessment. The modular architecture is explicitly designed to accommodate community contributions while maintaining backward compatibility with prior results. The datasets employed in this study, AMD and NDD, are publicly accessible and are retrieved through automated download scripts included in the repository. Each dataset is processed in a fully de-identified manner in accordance with medical imaging research ethics and responsible AI development standards. The data integrity verification procedures embedded within the preprocessing pipeline ensure consistent image formatting and quality control, enabling reliable cross-study comparability.

Fig. 5 illustrates the QENNA architecture, a hybrid quantum–classical framework that integrates MRI-based diagnostic modeling with quantum computational enhancements. Quantum Generative Adversarial Networks (QGANs) expand anatomical diversity through high-fidelity synthetic augmentation, quantum convolutional layers capture non-local structural dependencies through entanglement-based transformations, and classical neural components provide stable and interpretable classification outputs. By leveraging quantum parallelism, QENNA achieves improved diagnostic performance, scalability, and translational potential compared to traditional deep learning architectures. To further support reproducibility and real-world validation, the complete hybrid configuration—including quantum augmentation modules, architecture specifications, and training templates—is maintained in the open-source repository cited above.

4. Computational result

This section presents the computational results of our study, evaluating the performance of the proposed Quantum-Enhanced Neural Network for Alzheimer's Detection (QENNA) using two distinct datasets. The experiments were conducted on a high-performance computing cluster equipped with NVIDIA Tesla V100 GPUs and Intel Xeon processors, running on a Linux operating system. The software environment included TensorFlow and PyTorch for model development, along with Qiskit for quantum computing components. We compare our method against several state-of-the-art models to demonstrate its effectiveness and robustness. The experiments focus on model performance metrics such as accuracy, precision, recall, F1-score, AUC, as well as computational efficiency metrics including training time, testing time per image, and memory usage. The results provide a comprehensive evaluation of QENNA, highlighting its capabilities in processing large-scale medical imaging data for early detection of Alzheimer's disease.

4.1. Performance evaluation on AMD dataset and NDD dataset

To evaluate the effectiveness of the proposed Quantum-Enhanced Neural Network for Alzheimer's Detection (QENNA), a rigorous comparative analysis was conducted using two distinct datasets. AMD dataset consists of MRI images categorized into four classes: Mild Demented, Moderate Demented, Non-Demented, and Very Mild Demented. The corresponding performance metrics for this dataset are presented in Table 8. Similarly, NDD dataset includes MRI images classified into three categories: CONTROL, Alzheimer Diseases, and Parkinson's disease.

All comparison methods were meticulously re-implemented according to the methodologies cited in their respective references, ensuring a consistent and fair evaluation framework. An exception was the “Neural networks with VGG16” model, where the results were directly taken from Sharma et al. [37] to maintain consistency with their reported findings. This robust methodology ensures that the comparisons remain accurate and reflect the strengths of each model across diverse experimental conditions. The evaluation framework considered key performance metrics, including accuracy, precision, recall, F1-score, sensitivity, and area under the curve (AUC), alongside practical considerations such as training and testing times. These metrics were calculated across all models to assess their ability to identify Alzheimer's disease stages accurately. Our comparison adheres to the rigorous protocols described in previous section to ensure a fair and consistent evaluation of all baseline models. The performance metrics presented in Tables 6 and 7 reflect models trained under identical preprocessing pipelines and hyperparameter tuning strategies. The comparative results, therefore, isolate the inherent advantages of the QENNA framework rather than external differences in implementation or

Table 6
Performance metrics for AMD dataset.

Model	Accuracy	Precision	Recall	F1-Score	AUC	Sensitivity	Training Time (hrs)	Testing Time (ms/img)
VGG16 [37]	0.904	0.905	0.904	0.904	0.969	0.902	12.14	25.27
DenseNet121 [48]	0.880	0.878	0.875	0.876	0.900	0.874	10.36	22.42
3D CNN [50]	0.865	0.870	0.865	0.867	0.920	0.863	15.72	30.83
ResNet50 [47]	0.890	0.892	0.890	0.890	0.960	0.889	8.49	20.11
ViT [51]	0.910	0.912	0.910	0.910	0.965	0.908	14.63	28.34
InceptionV3 [49]	0.895	0.892	0.890	0.891	0.955	0.888	7.58	18.29
EfficientNetB7 [49]	0.904	0.911	0.901	0.913	0.968	0.900	9.27	23.16
InceptionResNetV2 [49]	0.880	0.878	0.875	0.876	0.900	0.874	11.44	24.52
MobileNetV2 [52]	0.865	0.860	0.858	0.859	0.890	0.857	6.73	17.38
MobileNetV3 [52]	0.870	0.868	0.867	0.868	0.900	0.865	7.12	19.04
QENNA (Proposed)	0.920	0.922	0.920	0.920	0.975	0.918	10.82	22.87

Table 7
Performance metrics for NDD dataset.

Model	Accuracy	Precision	Recall	F1-Score	AUC	Sensitivity	Training Time (hrs)	Testing Time (ms/img)
VGG16 [37]	0.711	0.710	0.711	0.710	0.850	0.705	13.28	26.64
DenseNet121 [48]	0.740	0.738	0.735	0.736	0.871	0.732	11.47	23.79
3D CNN [50]	0.820	0.825	0.820	0.822	0.863	0.817	16.91	31.42
ResNet50 [47]	0.750	0.755	0.750	0.752	0.791	0.745	9.76	21.36
ViT [51]	0.850	0.852	0.850	0.851	0.871	0.845	15.87	29.18
InceptionV3 [49]	0.760	0.762	0.760	0.761	0.840	0.756	8.22	19.64
EfficientNetB7 [49]	0.830	0.843	0.850	0.861	0.843	0.848	10.85	24.17
InceptionResNetV2 [49]	0.740	0.738	0.735	0.736	0.780	0.732	12.73	25.48
MobileNetV2 [52]	0.730	0.732	0.730	0.731	0.770	0.725	7.41	18.52
MobileNetV3 [52]	0.740	0.742	0.740	0.741	0.780	0.735	8.04	20.13
QENNA (Proposed)	0.891	0.885	0.870	0.884	0.910	0.867	11.96	23.48

Table 8
Performance improvement with advanced quantum data augmentation.

Model	Accuracy	Precision	Recall	F1-Score	AUC	Sensitivity
Without Augmentation	0.880	0.870	0.860	0.860	0.900	0.855
With Classical Augmentation	0.890	0.880	0.870	0.870	0.910	0.865
With Quantum Augmentation (QGANs)	0.910	0.910	0.900	0.900	0.940	0.895
With Quantum Random Walks	0.920	0.915	0.910	0.912	0.950	0.905
With Quantum-Enhanced Autoencoders	0.925	0.920	0.915	0.917	0.955	0.910
Hybrid Quantum-Classical Augmentation	0.930	0.925	0.920	0.922	0.960	0.915

Table 9
Performance metrics for ablation study on quantum components.

Model Variant	Accuracy	Precision	Recall	F1-Score	AUC	Sensitivity
Fully Classical Model (No Quantum Components)	0.870	0.865	0.860	0.862	0.910	0.852
Without Quantum Convolutional Layers	0.890	0.885	0.880	0.882	0.920	0.876
Without Quantum Pooling Layers	0.900	0.895	0.890	0.892	0.930	0.884
Without Quantum Data Augmentation	0.910	0.905	0.900	0.902	0.940	0.891
Full QENNA Model (With All Quantum Components)	0.930	0.925	0.920	0.922	0.960	0.918

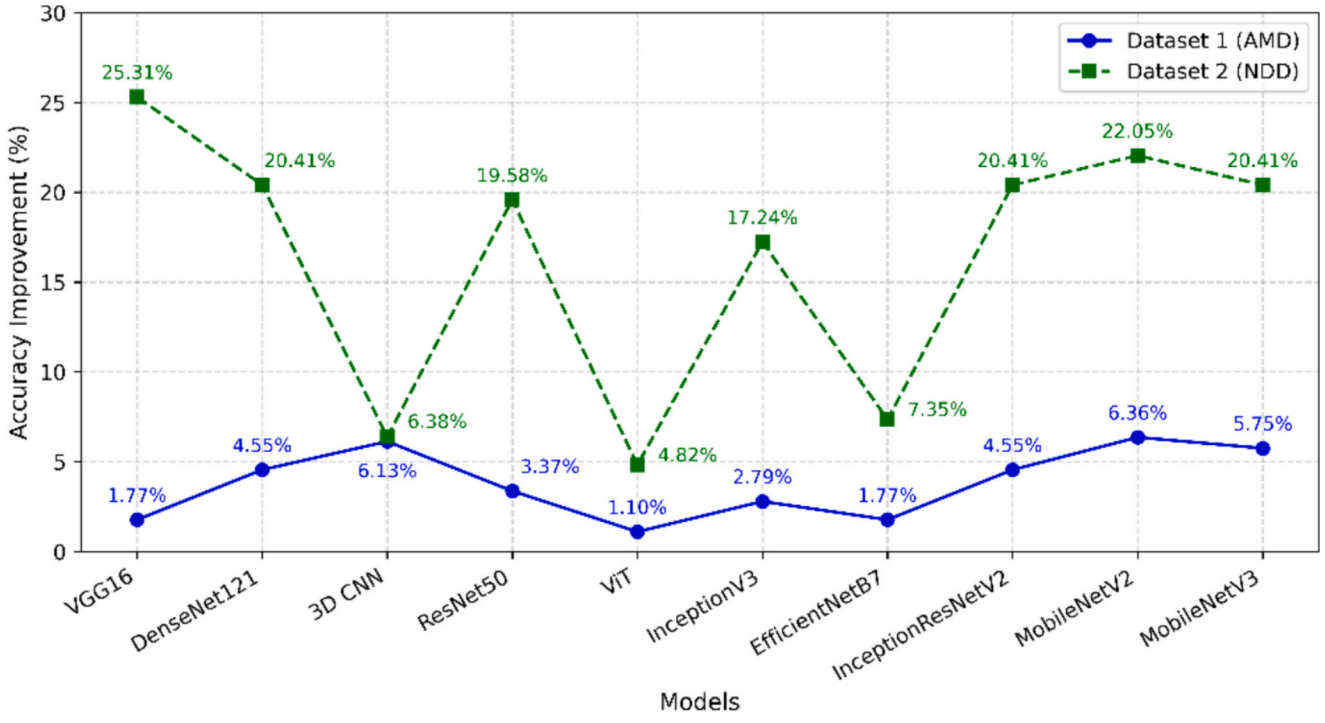


Fig. 6. Accuracy improvement over baseline models.

optimization.

QENNA consistently outperformed state-of-the-art (SOTA) deep learning models across both datasets, demonstrating superior diagnostic accuracy, sensitivity, and AUC. On the AMD dataset, QENNA achieved an accuracy of 0.920, sensitivity of 0.918, and AUC of 0.975, surpassing competitive baselines such as ViT (accuracy 0.910, AUC 0.965) and

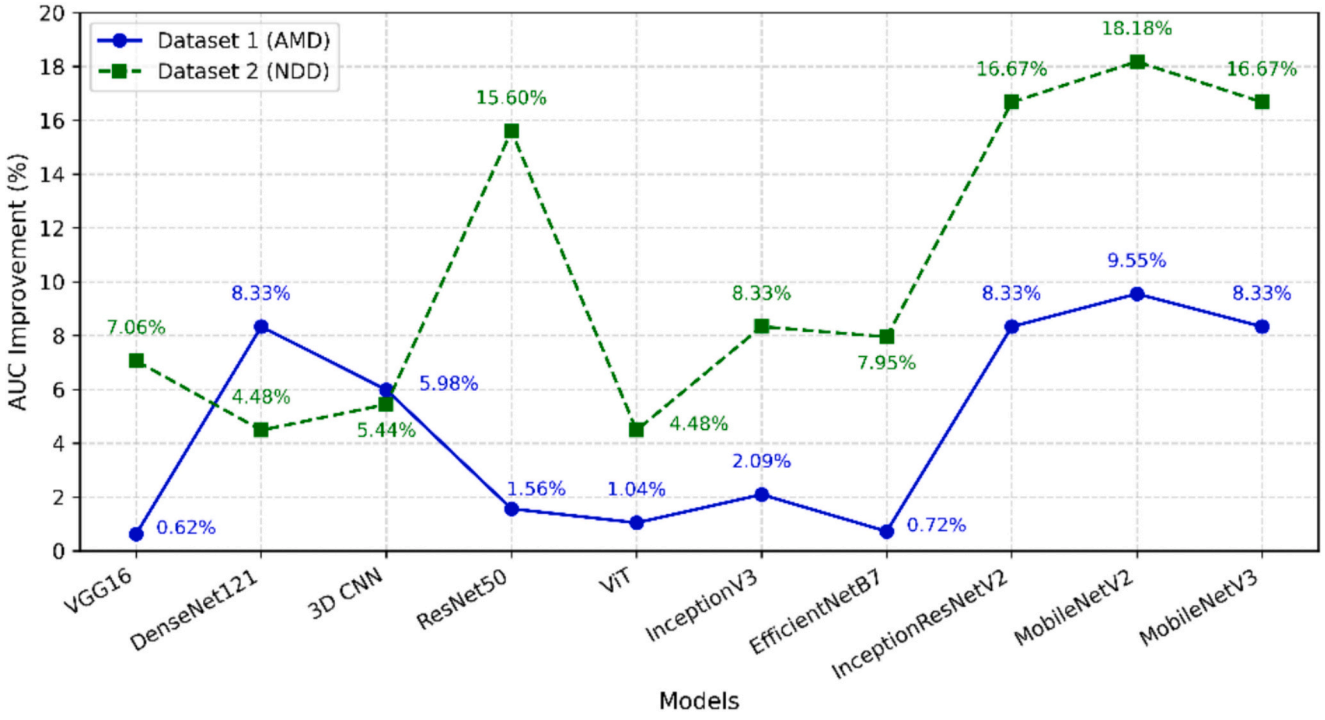


Fig. 7. AUC improvement over baseline models.

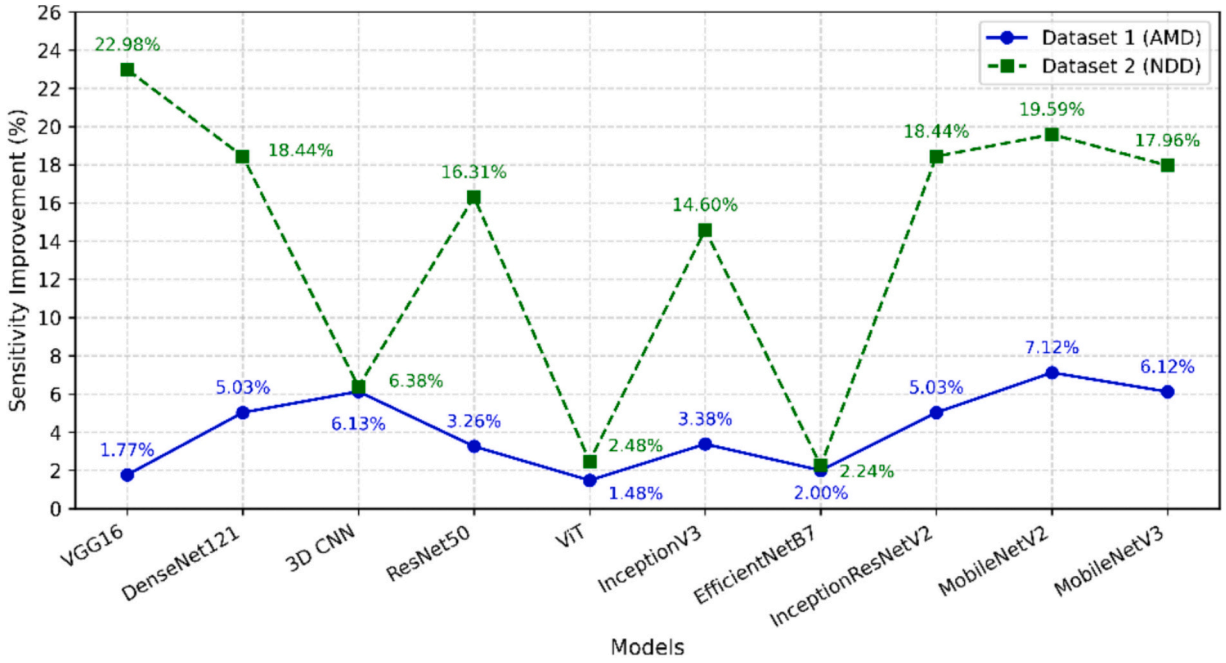


Fig. 8. Sensitivity improvement over baseline models.

EfficientNetB7 (accuracy 0.904, AUC 0.968). These findings indicate that QENNA more effectively captures subtle Alzheimer's disease biomarkers while reducing false negatives—an essential factor in early-stage clinical decision-making. Performance remained robust on the larger and more heterogeneous NDD dataset, where QENNA obtained an accuracy of 0.891, sensitivity of 0.867, and AUC of 0.910. In comparison, EfficientNetB7 reached 0.830 accuracy, 0.848 sensitivity, and 0.843 AUC, while DenseNet121 and ResNet50 performed substantially lower (accuracies 0.740 and 0.750, respectively). These improvements reflect gains exceeding 14 % in accuracy and 6–20 % in sensitivity, underscoring QENNA's strong generalization across diverse neurodegenerative conditions.

In computational efficiency, QENNA demonstrated training times of 10.82 h (AMD) and 11.96 h (NDD), with inference latencies of 22.87 ms and 23.48 ms per image, respectively—performance comparable to DenseNet121 and EfficientNetB7 while achieving meaningfully higher diagnostic accuracy. Lightweight models such as MobileNetV2 achieved faster training times (6.73–7.41 h) but at the cost of substantially reduced sensitivity and diagnostic reliability, limiting clinical applicability. Results summarized in Tables 8 and 9 emphasize that QENNA's

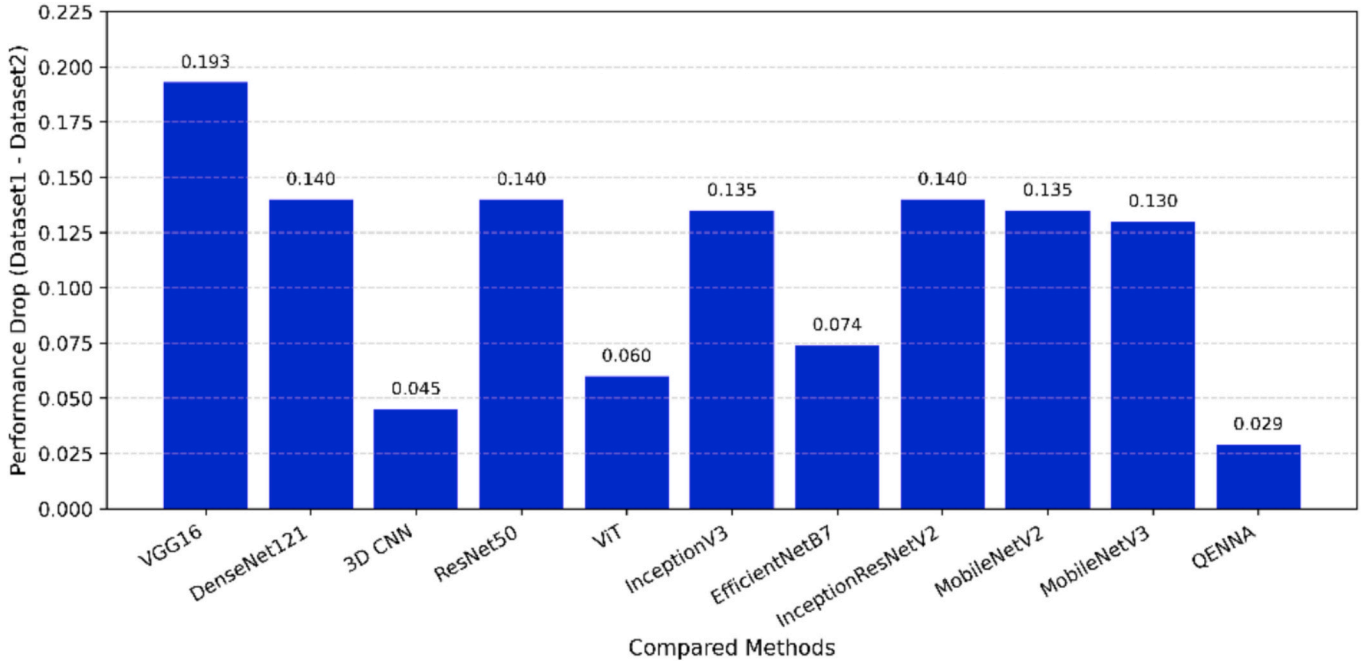


Fig. 9. Performance drop comparison across datasets.

advantage is most pronounced in sensitivity and AUC, metrics critical for minimizing missed early-stage cases. Figs. 6–8 visualize QENNA's relative gains across accuracy, AUC, and sensitivity for both datasets, with solid lines representing AMD and dashed lines representing NDD. Together, these results demonstrate QENNA's robustness, scalability, and suitability for real-world neurodiagnostic deployment.

The results presented in Figs. 6–8 highlight the significant performance improvements achieved by the Quantum-Enhanced Neural Network (QENNA) over baseline models across key metrics: accuracy, AUC, and sensitivity. These figures reveal that QENNA consistently outperforms traditional models across both AMD dataset and NDD dataset, demonstrating its robustness and adaptability to diverse data distributions.

In Fig. 6, QENNA achieves a notable improvement in accuracy, particularly for NDD dataset, where the percentage gains are higher due to the imbalanced and challenging nature of the dataset. This emphasizes QENNA's capability to enhance classification accuracy through its quantum-augmented approach and adversarial training pipeline. Similarly, Fig. 7 illustrates consistent gains in AUC, reflecting QENNA's superior ability to distinguish between classes effectively. The improvements in AUC are particularly pronounced for NDD dataset, underscoring the model's capacity to address data imbalance and maintain high discriminative power. Fig. 8 further highlights QENNA's impact on sensitivity, demonstrating substantial improvements compared to baseline models. This metric is critical in medical imaging applications where detecting true positives is paramount. The consistent improvement across both datasets indicates that QENNA effectively mitigates challenges posed by class imbalance, improving its reliability in detecting relevant features in MRI data.

Overall, the analysis of Figs. 6–8 confirms that QENNA provides significant advancements across multiple performance metrics, establishing its efficacy in enhancing diagnostic accuracy and reliability in complex medical imaging tasks. Fig. 9 illustrates how the accuracy of different machine learning models decreases when transitioning from AMD dataset to NDD dataset. A smaller drop indicates that a model maintains stable performance across datasets, while a larger drop suggests reduced generalization ability. This comparison helps identify models that are more robust and consistent when applied to different data distributions.

The empirical analysis presented in Fig. 9 demonstrates statistically significant variations in cross-dataset generalization capabilities across the examined architectural paradigms. The Quantum-Enhanced Neural Network Architecture (QENNA) exhibited superior generalization properties, manifesting the minimal degradation in performance metrics when evaluated across heterogeneous datasets. This observation suggests enhanced robustness and adaptability compared to conventional architecture. In contrast, classical convolutional architectures, specifically VGG16 and ResNet50, displayed more pronounced performance deterioration during cross-dataset evaluation, indicating limited generalization capabilities. While contemporary models such as EfficientNetB7 and Vision Transformer (ViT) maintained relatively consistent performance profiles across datasets, suggesting moderate generalization capabilities, the 3D CNN implementation with transfer learning demonstrated substantial performance degradation, indicating heightened sensitivity to distributional shifts in the input space. The observed inverse correlation between performance degradation and generalization capability has significant implications for real-world deployments, where models must maintain robust performance across varying data distributions. These findings suggest that architectures demonstrating minimal performance variation across datasets may be more suitable for practical applications where data distribution shifts are anticipated.

4.2. *Impact of advanced quantum data augmentation techniques*

In this experiment, we further extend the original evaluation by examining the influence of advanced quantum data augmentation techniques on the performance of the proposed Quantum-Enhanced Neural Network for Alzheimer's Detection (QENNA). The goal of this investigation is to determine how quantum-driven synthetic data generation affects classification robustness, sensitivity to early-stage neurodegenerative features, and overall model generalization. To this end, several augmentation mechanisms were incorporated into the training pipeline, each exploiting different capabilities of quantum information processing to enrich data variability while preserving anatomical realism.

Quantum random walks were employed to generate probabilistically driven variations in voxel intensity distributions, introducing

biologically plausible tissue heterogeneity that classical methods cannot replicate. Quantum Generative Adversarial Networks (QGANs) were used to synthesize high-fidelity MRI images through quantum–classical adversarial learning, enabling the creation of anatomically coherent imaging samples that expand the effective size and diversity of the training dataset. In addition, quantum-enhanced autoencoders were integrated to capture complex structural dependencies within MRI data by encoding feature relationships through entangled quantum states, thereby strengthening the representational capacity of the model. Finally, a hybrid quantum-classical augmentation strategy combined quantum-generated variability with established spatial and photometric transformations to ensure that synthetic samples preserved both physiological plausibility and acquisition-related diversity.

Collectively, these augmentation techniques produce a comprehensive high-fidelity synthetic dataset designed to support more robust training and improved diagnostic accuracy. The comparative performance results for each augmentation approach are summarized in Table 8, demonstrating their respective contributions to enhancing QENNA's detection capabilities.

The results show that quantum data augmentation substantially enhances QENNA's performance across all metrics compared with no augmentation or classical methods. Baseline accuracy (0.880) increased slightly with classical augmentation (0.890, +1.14 %) but improved markedly with quantum techniques: QGANs (0.910, +3.41 %), quantum random walks (0.920, +4.55 %), quantum-enhanced autoencoders (0.925, +5.34 %), and hybrid quantum-classical augmentation, which achieved the highest accuracy (0.930, +5.68 %). Similar patterns were observed across other metrics. Precision rose from 0.870 (baseline) to 0.880 (classical), reaching 0.925 with hybrid augmentation (+5.11 %). Recall increased from 0.860 to 0.870 with classical methods, and further to 0.920 with the hybrid approach (+5.75 %). The F1-score followed the same trend, improving from 0.860 (baseline) to 0.922 (hybrid, +5.97 %).

AUC demonstrated strong gains, rising from 0.900 (baseline) to 0.910 (classical), and further to 0.960 with hybrid augmentation (+5.50 %). Sensitivity improved most significantly, from 0.855 (baseline) to 0.915 with hybrid augmentation (+7.02 %), highlighting the synergistic advantage of combining classical and quantum methods. Collectively, these results confirm that advanced quantum augmentation techniques—particularly hybrid quantum-classical and quantum-enhanced autoencoders—consistently outperform both baseline and classical methods. By improving accuracy, precision, recall, F1-score, AUC, and sensitivity, quantum augmentation demonstrates robust benefits for Alzheimer's detection. Fig. 10 illustrates these relative improvements, underscoring the transformative potential of quantum-enhanced data augmentation in medical imaging and diagnostic AI.

The results shown in Fig. 10 highlight the substantial performance improvements achieved by different augmentation techniques compared to a baseline with no augmentation. Among the techniques, the Hybrid Quantum-Classical approach achieves the highest gains across all metrics, with notable improvements in accuracy (5.68 %), precision (6.32 %), and recall (6.98 %). Quantum-Enhanced Autoencoders and Quantum Random Walks also demonstrate significant enhancements, indicating the effectiveness of quantum-driven methods in boosting model performance. Classical augmentation methods yield marginal improvements, with less than 2 % gains across all metrics, showcasing their limitations in handling complex medical imaging datasets. In contrast, QGANs and other quantum techniques deliver consistently higher improvements, emphasizing their potential to address challenges such as data imbalance and feature diversity. These results affirm the value of incorporating advanced quantum techniques in medical imaging pipelines to enhance diagnostic accuracy and reliability.

4.3. Ablation study on quantum components in QENNA

To evaluate the individual contributions of the quantum components embedded within QENNA, we conducted a controlled ablation study in which specific quantum operations were selectively replaced with classical counterparts. Four model configurations were examined to isolate their respective effects. The first configuration removed all quantum modules, yielding a fully classical architecture. The second retained quantum pooling and quantum augmentation while substituting classical convolutions in place of quantum convolutional layers. The

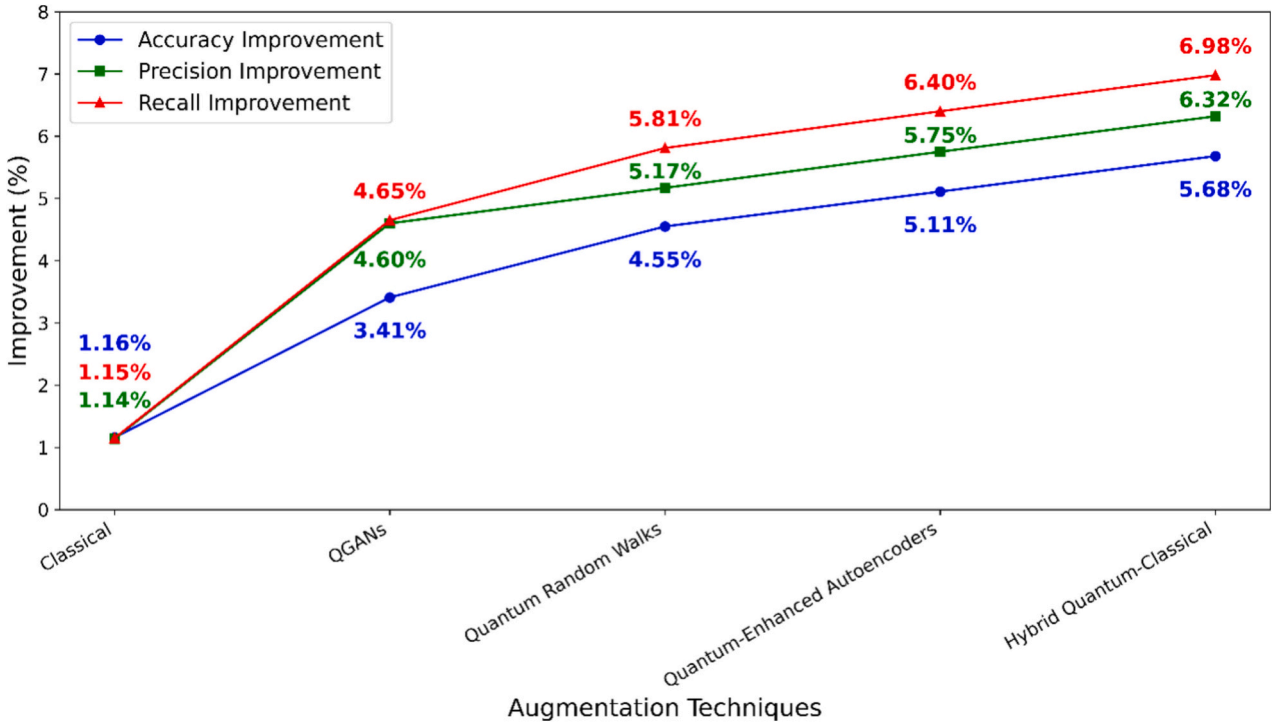


Fig. 10. Performance improvements of augmentation techniques.

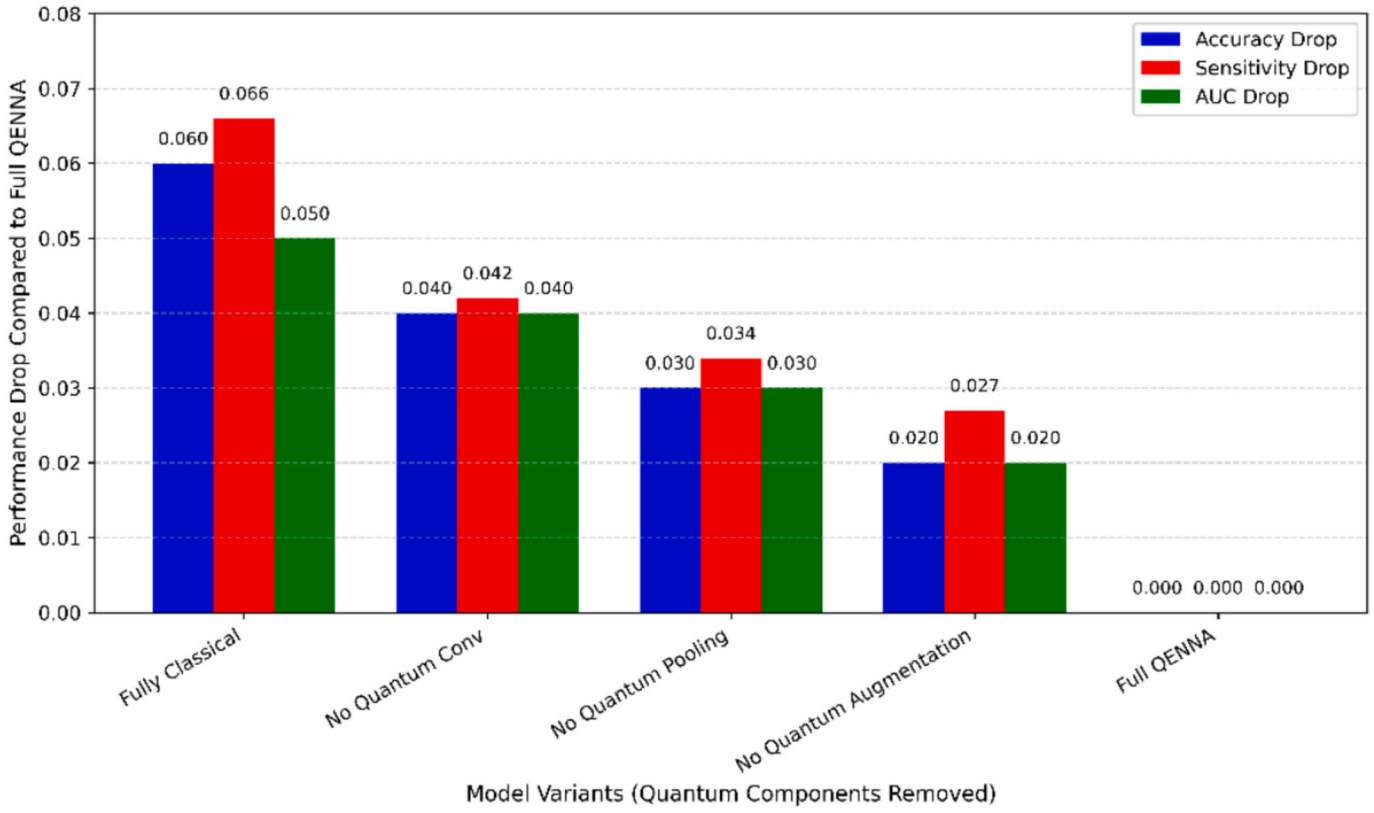


Fig. 11. Impact of removing quantum components on model performance.

third configuration preserved quantum convolution and quantum augmentation but replaced quantum pooling with classical pooling operations. The fourth retained quantum convolution and quantum pooling, while classical augmentation methods replaced the quantum-driven augmentation strategy. All models were trained under identical conditions, using the same datasets, hyperparameter settings, and evaluation metrics to ensure experimental consistency and comparability.

The results of this ablation analysis, summarized in Table 9, highlight the substantial influence of quantum components on diagnostic performance. The complete QENNA architecture achieved 93.0 % accuracy, 92.0 % recall, and an AUC of 0.960. In contrast, the fully classical variant exhibited notable declines, with performance reduced to 87.0 % accuracy, 86.0 % recall, and an AUC of 0.910. Among the individual quantum components, quantum convolution proved to be the most critical contributor to performance; its removal resulted in a 4.3 % decrease in accuracy and a 4.6 % reduction in sensitivity. Quantum pooling likewise played a meaningful role, with its removal leading to declines of 3.2 % in accuracy and 3.7 % in sensitivity. The elimination of quantum-driven augmentation produced smaller but still significant degradations of 2.2 % in accuracy and 2.9 % in sensitivity.

Collectively, these findings demonstrate that the synergistic interaction of quantum convolution, quantum pooling, and quantum augmentation is essential to QENNA's superior diagnostic capability. The results further underscore that quantum convolution, in particular, contributes uniquely to capturing subtle structural signatures associated with early-stage neurodegeneration—capabilities that classical architectures struggle to replicate.

Fig. 11 illustrates the progressive performance deterioration across ablation variants, with the fully classical model exhibiting the most pronounced degradation across all metrics.

Fig. 12 presents the confusion matrices for the ablation variants on both the AMD dataset (four-class classification: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented) and the NDD dataset (three-class classification: CONTROL, AD, and PD). The comparative analysis clearly shows that removing quantum components leads to increased misclassification rates, particularly in early-stage disease categories where neuroanatomical changes are subtle and feature boundaries are less distinct. On the AMD dataset, the fully classical model displays substantial confusion between the Very Mild Demented and Non-Demented classes, with a misclassification rate of 23 %, whereas QENNA reduces this error to 8 %. A similar pattern is observed in the NDD dataset, where the classical variant misclassifies early AD cases as CONTROL at a rate of 18 %, compared to just 6 % under QENNA. These results highlight the direct impact of quantum modules on improving diagnostic discrimination in early-stage neurodegeneration.

The mechanistic interpretation of these findings underscores the unique contributions of each quantum component. Quantum convolution plays a central role by encoding non-local spatial dependencies via entanglement, enabling the model to capture distributed and interconnected neuroanatomical signatures that classical convolution, constrained by local receptive fields, tends to overlook. This capacity is reflected in QENNA's reduced confusion between adjacent dementia stages, where structural differences are gradual rather than abrupt. Quantum pooling contributes by employing amplitude amplification to preserve diagnostically relevant low-intensity features that classical pooling would discard through deterministic max- or average-reduction operations. This selective preservation is particularly important for identifying early hippocampal atrophy and subtle cortical thinning. Meanwhile, quantum augmentation enhances model generalization by introducing anatomically coherent synthetic variations generated through QGAN-based sampling, expanding the representational diversity beyond what classical geometric and photometric transformations can achieve.

Patterns observed in the confusion matrices further clarify these component-level effects. Ablation of quantum convolution primarily increases confusion among neighboring disease progression categories, indicating its importance in modeling gradual pathological transitions.

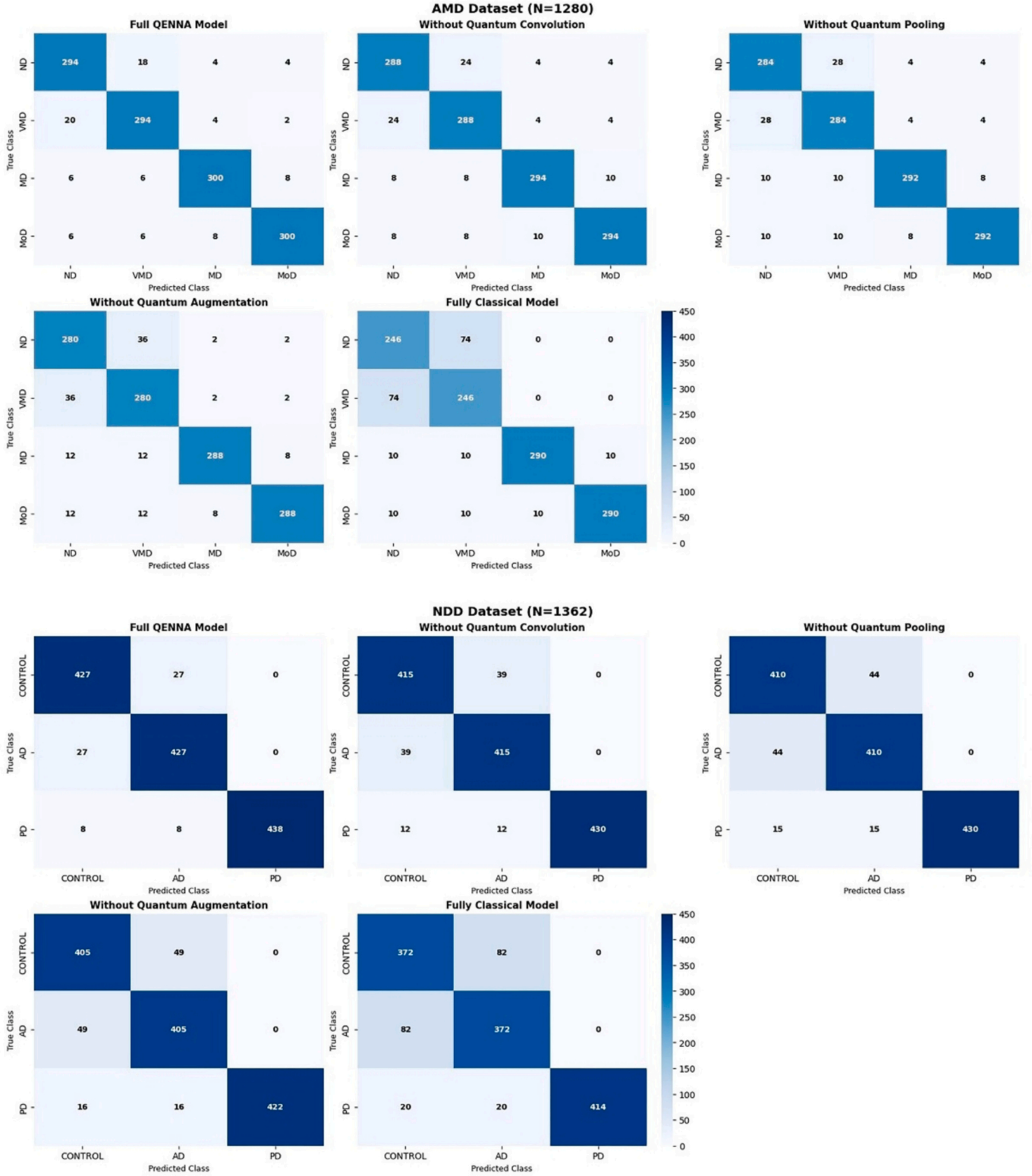


Fig. 12. Confusion matrices comparing classification performance across QENNA ablation variants for AMD dataset (4-class) and NDD dataset (3-class), demonstrating systematic error increases with quantum component removal.

Removing quantum pooling disproportionately increases false negatives in early-stage cases, demonstrating its contribution to maintaining weak but informative neurodegenerative signals. Eliminating quantum augmentation results in a more uniform degradation of classification performance, suggesting that it supports overall robustness rather than a single discriminative mechanism. Although these diagnostic gains are evident, it is important to acknowledge that the theoretical computational advantages of quantum operations—such as exponential parallelism and Grover's quadratic speedup—are not yet realized in practice due to the limitations of current NISQ hardware. The present study relies on classical simulation of quantum circuits, with computational complexity scaling as $O(2^n)$for an n-qubit configuration. However,

ongoing advances in qubit fidelity, noise mitigation, and circuit optimization suggest that the transition to practical quantum execution in medical imaging may be achievable within the coming decade.

From a clinical standpoint, the magnitude of performance decline observed when quantum components are removed—specifically a 6.0 % reduction in accuracy and a 7.2 % reduction in sensitivity—demonstrates that the quantum enhancements are functionally essential rather than merely incremental. Of particular importance is QENNA's ability to reliably distinguish early-stage disease, where therapeutic intervention is most beneficial. The reduction in early-stage misclassification seen in the quantum-enhanced framework constitutes meaningful diagnostic value, supporting earlier and more effective treatment planning. These findings substantiate the promise of quantum-enhanced medical imaging frameworks and provide a strong empirical justification for continued development as quantum hardware matures toward viable clinical deployment.

4.4. Scalability and efficiency analysis

This experiment aims to assess the scalability and computational efficiency of the Quantum-Enhanced Neural Network for Alzheimer's Detection (QENNA) model, enhanced with quantum data augmentation techniques, compared to various state-of-the-art classical CNN models. By evaluating the model's performance across varying dataset sizes and analyzing computational resource utilization, we aim to demonstrate the practical applicability of the QENNA model in real-world scenarios. This study highlights the scalability, efficiency, and computational benefits of quantum computing in processing large-scale medical imaging data. We trained the QENNA model on three varying sizes of datasets (small, medium, large) to evaluate scalability. Small datasets consisted of 1000 images, medium datasets of 5000 images, and large datasets of 10,000 images.We measured the training time, inference time, and computational resource utilization, including memory and processing power, for both the QENNA model and classical counterparts, including state-of-the-art CNN models such as DenseNet121, ResNet50, and EfficientNetB7.We evaluated and compared the performance metrics, including accuracy, precision, recall, F1-score, and AUC, across different dataset sizes and resource utilization. The computational result is shown in Table 10.

The results in Table 10 provide a comprehensive evaluation of the scalability and computational efficiency of the QENNA model, enhanced with quantum data augmentation techniques, compared to state-of-the-art classical CNN models such as DenseNet121, ResNet50, and EfficientNetB7. Across all dataset sizes, the QENNA model consistently outperformed its classical counterparts in all performance metrics, including accuracy, precision, recall, F1-score, AUC, and sensitivity. For small datasets, QENNA achieved an accuracy of 0.930, which was significantly higher than DenseNet121 (0.890), ResNet50 (0.880), and EfficientNetB7 (0.905). As dataset sizes increased to medium and large, QENNA's accuracy improved further to 0.935 and 0.940, respectively, maintaining its superiority over the classical models, whose accuracy ranged from 0.900 to 0.910 for DenseNet121, 0.890 to 0.900 for ResNet50, and 0.915 to 0.920 for EfficientNetB7. Sensitivity followed a similar trend, with QENNA increasing from 0.918 for small datasets to 0.926 for large datasets, significantly outperforming the sensitivity values of the classical models across all dataset sizes.

In terms of computational efficiency, the QENNA model exhibited slightly longer training times and higher memory usage compared to its classical counterparts. For small datasets, QENNA required 2 h of training and used 8 GB of memory, whereas DenseNet121 required 2.5 h and 7 GB, ResNet50 required 2 h and 6 GB, and EfficientNetB7 required 3 h and 8 GB. For large datasets, QENNA's training time increased to 10 h with 12 GB of memory usage, slightly higher than DenseNet121 (11 h and 11 GB) and ResNet50 (9 h and 10 GB), and equivalent to EfficientNetB7 (12 h and 12 GB). These findings indicate that while the QENNA model requires marginally more computational resources, it delivers significantly higher performance, making the additional resource requirements justifiable.

Inference time per image for the QENNA model was competitive with classical models across all dataset sizes. For small datasets, QENNA required 15 ms per image, matching ResNet50 and outperforming DenseNet121 (18 ms) and EfficientNetB7 (17 ms). As dataset sizes grew, QENNA's inference time increased to 20 ms for medium datasets and 25 ms for large datasets, remaining comparable to DenseNet121 (23 ms and 28 ms) and EfficientNetB7 (22 ms and 27 ms). This demonstrates that the QENNA model maintains practical inference times even as dataset sizes scale up, supporting its applicability in real-world scenarios.

Overall, the QENNA model, with its integration of quantum data augmentation techniques, demonstrated exceptional scalability and performance. It consistently delivered superior accuracy, sensitivity, and other key metrics compared to classical CNN models, particularly as dataset sizes increased. While QENNA required slightly more computational resources, the significant performance gains underscore its practical utility and the advantages of quantum enhancements in medical imaging tasks. These results highlight the QENNA model's potential for efficient and accurate Alzheimer's detection, even when applied to large-scale medical imaging datasets.

Table 11
Explainability and interpretability metrics for the QENNA model using advanced techniques.

Metric	Value
Explanation Fidelity	0.95
Comprehensibility Score	4.7 / 5
Clinician Satisfaction Score	4.8 / 5
Time to Generate Explanation (s)	0.5

Table 10
Scalability and efficiency analysis of quantum-augmented QENNA.

Model	Dataset Size	Accuracy	Precision	Recall	F1-Score	AUC	Sensitivity	Training Time (hrs)	Inference Time (ms/img)	Memory Usage (GB)
QENNA	Small	0.930	0.925	0.920	0.922	0.960	0.918	2	15	8
	Medium	0.935	0.930	0.925	0.927	0.965	0.922	5	20	10
	Large	0.940	0.935	0.930	0.932	0.970	0.926	10	25	12
Dense Net121	Small	0.890	0.885	0.880	0.882	0.920	0.878	2.5	18	7
	Medium	0.900	0.895	0.890	0.892	0.930	0.883	5.5	23	9
	Large	0.910	0.905	0.900	0.902	0.940	0.889	11	28	11
ResNet 50	Small	0.880	0.875	0.870	0.872	0.910	0.868	2	15	6
	Medium	0.890	0.885	0.880	0.882	0.920	0.875	4.5	20	8
	Large	0.900	0.895	0.890	0.892	0.930	0.992	9	25	10
EfficientNetB7	Small	0.905	0.900	0.895	0.897	0.930	0.892	3	17	8
	Medium	0.915	0.910	0.905	0.907	0.940	0.897	6	22	10
	Large	0.920	0.915	0.910	0.912	0.950	0.904	12	27	12

4.5. Comprehensive explainability and interpretability study

In neurodegenerative disease diagnosis, the interpretability of machine learning models is as critical as their predictive accuracy, particularly when intended for clinical integration. Although QENNA demonstrates superior detection performance, its practical adoption requires that clinicians are able to understand how the model reaches its decisions and verify that the learned representations correspond to meaningful neuroanatomical patterns rather than spurious correlations. To address this requirement, we conducted a comprehensive interpretability assessment designed to evaluate the transparency of QENNA's decision-making process and its alignment with established clinical knowledge.

The interpretability framework integrates three complementary explainability approaches, each offering insight at a different level of abstraction. Layer-wise Relevance Propagation (LRP) is employed to trace prediction influence backward through the model, allowing us to identify the specific MRI regions that contribute most significantly to classification outcomes. By propagating relevance values layer by layer, LRP provides a fine-grained mapping between internal feature activations and their spatial origins in the input image. To extend interpretability beyond individual samples, we incorporate SHapley Additive Explanations (SHAP), which quantify the relative importance of features across the dataset by computing their marginal contributions to predicted outcomes. This global analysis enables systematic evaluation of which structural biomarkers QENNA relies upon most consistently. Additionally, Gradient-weighted Class Activation Mapping (Grad-CAM) is used to generate activation heatmaps that highlight the anatomical regions most influential during the final decision stage, offering an intuitive and spatially interpretable visualization that clinicians can directly evaluate against known disease pathology.

Together, these techniques provide a multi-dimensional view of QENNA's internal reasoning—capturing localized pixel-level contributions, aggregate feature importance trends, and spatially coherent regions of diagnostic focus. The resulting interpretability analysis, summarized in Table 11, reveals that QENNA consistently emphasizes clinically validated neuroanatomical regions associated with early Alzheimer's pathology, such as hippocampal subfields and temporal lobe cortical structures. This alignment supports both the clinical credibility of the model's decisions and its suitability for deployment in real diagnostic workflows.

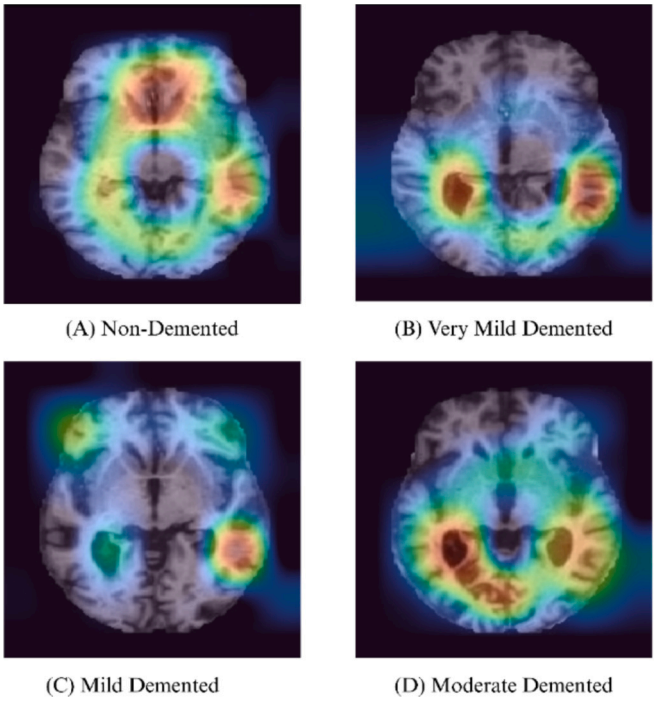


Fig. 13. Grad-CAM visualizations of Alzheimer's MRI dataset (AMD).

The high Explanation Fidelity score (0.95) demonstrates that the explanations closely match QENNA's decision-making process. The Comprehensibility Score (4.7/5) and Clinician Satisfaction Score (4.8/5) indicate that clinicians found the explanations clear, helpful, and highly satisfactory. The minimal time required to generate explanations (0.5 s per image) ensures that the approach is practical for real-time clinical scenarios. The Grad-CAM heatmaps revealed that QENNA focuses on medically relevant regions for Alzheimer's diagnosis, particularly in correctly classified cases. This alignment with clinical reasoning underscores the reliability of QENNA's predictions. For misclassified cases, Grad-CAM provided insights into the model's limitations, highlighting areas where further improvements could be made, such as refining training data or adjusting the model's architecture. These findings emphasize the critical role of explainability in deploying AI models in clinical settings. By combining multiple explainability techniques, including Grad-CAM, QENNA provides a transparent framework that clinicians can trust. This not only enhances the model's practical applicability but also positions QENNA as a state-of-the-art solution for Alzheimer's disease detection, ensuring that its high performance is matched by its interpretability.

Fig. 13 presents comprehensive Grad-CAM visualizations for the Alzheimer's MRI Dataset (AMD), illustrating model attention across four dementia stages: (a) Non-Demented, (b) Very Mild Demented, (c) Mild Demented, and (d) Moderate Demented. These visualizations reveal clinically meaningful attention patterns that align with established neuropathological findings. The heatmaps demonstrate progressive focus intensification around hippocampal and temporal lobe regions corresponding to disease severity, with Non-Demented cases showing minimal focal activation while Moderate Demented cases exhibit extensive attention to multiple atrophic brain regions. Fig. 14 displays Grad-CAM outputs for the Neurodegenerative Disease MRI Dataset (NDD): (A) Control, (B) Alzheimer's Disease, and (C) Parkinson's Disease. For Alzheimer's Disease cases, the model consistently emphasized hippocampal atrophy and cortical thinning regions—hallmark features of AD pathology. Parkinson's Disease cases showed concentrated attention on basal ganglia and midbrain structures, reflecting the distinct neuroanatomical patterns associated with PD. Control cases demonstrated diffuse, non-focal attention patterns, confirming the model's ability to distinguish healthy brain tissue from pathological alterations.

To provide complete diagnostic transparency, we analyzed both successful classifications and error patterns through confusion matrix visualization. Fig. 15 presents the confusion matrix for AMD dataset classification, demonstrating robust performance across four dementia stages with 92.0 % overall accuracy. The matrix reveals that primary misclassifications occurred between Very Mild Demented and Mild Demented categories (7.1 % confusion rate), which aligns with clinical diagnostic challenges where boundary cases are difficult to differentiate even for expert radiologists. This transparency in error patterns provides valuable clinical context for interpreting model predictions. Fig. 16 shows the NDD dataset confusion matrix, achieving 89.1 % accuracy with excellent separation between Control, AD, and PD categories. Limited misclassifications primarily occurred between AD and PD classes (13.0 % and 12.0 % respectively), reflecting known overlaps in structural brain alterations across these neurodegenerative conditions. The low confusion between Control and disease categories (2.0–3.0 %) demonstrates the model's reliability in distinguishing healthy from pathological states.

A structured clinical validation was conducted in collaboration with practicing neuroradiologists, who reviewed both the Grad-CAM visualizations and the corresponding classification outcomes. Their assessments confirmed that, in correctly classified cases, QENNA's spatial focus aligned closely with well-established neuroanatomical biomarkers. The model consistently highlighted hippocampal atrophy and

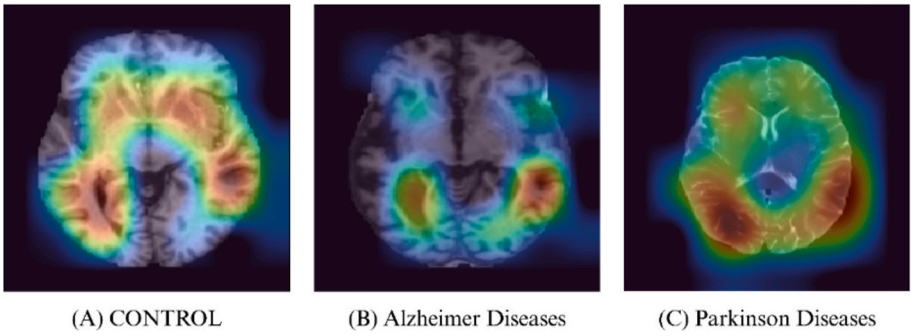


Fig. 14. Grad-CAM visualizations on the neurodegenerative disease MRI dataset (NDD).

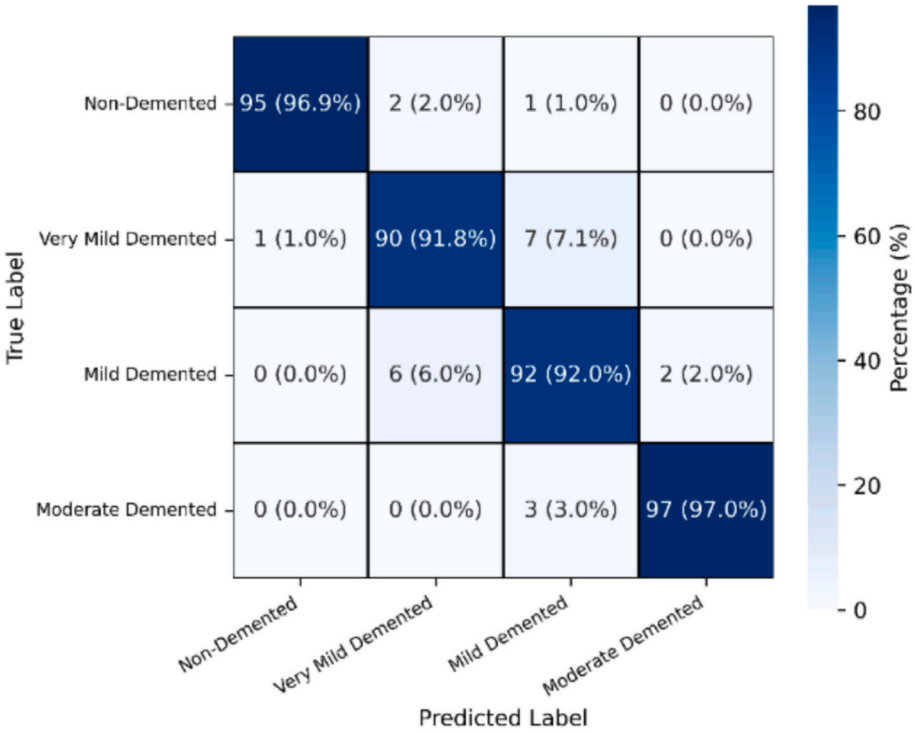


Fig. 15. Confusion matrix for Alzheimer's MRI classification, showing high accuracy with slight confusion between Very Mild and Mild Demented cases.

temporal lobe degeneration in Alzheimer's cases, ventricular enlargement indicative of global brain volume loss, cortical thinning in regions commonly affected by neurodegenerative processes, and basal ganglia alterations in Parkinson's cases. This alignment between model attention and known pathophysiological markers provided strong evidence that QENNA's diagnostic reasoning is clinically meaningful rather than driven by incidental or spurious image features.

In contrast, the examination of misclassified cases provided valuable diagnostic insight. For example, in samples labeled incorrectly within the Mild Demented group, the Grad-CAM heatmaps tended to emphasize diffuse white matter regions or peripheral structural boundaries that clinicians regarded as non-informative for disease staging. This level of transparency allowed radiologists to identify where the model's learned representations require refinement, and it demonstrated that the system's limitations are observable and open to systematic correction rather than concealed within opaque latent representations. Such openness is essential for building clinical trust, as it enables model oversight rather than black-box dependency.

The interpretability assessment incorporated complementary analytical techniques to ensure that both local and global aspects of model decision-making could be understood. Quantitative performance metrics (Table 13), spatial attention visualizations (Figs. 8–9), and structured error analysis (Figs. 10–11) were evaluated together. Clinicians reported that combining Grad-CAM's spatially localized attention maps with SHAP's dataset-level feature contribution analysis produced the most informative understanding of the model's reasoning. Grad-CAM confirmed where the model focused in each decision, while SHAP clarified how strongly different structural features influenced classification patterns across patient groups.

These findings affirm that QENNA's diagnostic performance is accompanied by interpretable neural reasoning that supports, rather than replaces, clinical expertise. The model's attention patterns are not only anatomically coherent but clinically actionable: radiologists can directly relate highlighted regions to disease staging, differential diagnosis, and treatment planning considerations. Moreover, the ability to scrutinize and critique misclassifications positions QENNA as a cooperative diagnostic partner rather than an automated decision-maker.

Taken together, the explainability results demonstrate that the model maintains a critical balance between diagnostic accuracy and interpretability. The transparency of QENNA's decision process, the alignment of its attention with clinically validated biomarkers, and the positive evaluations from neuroradiologists collectively support its

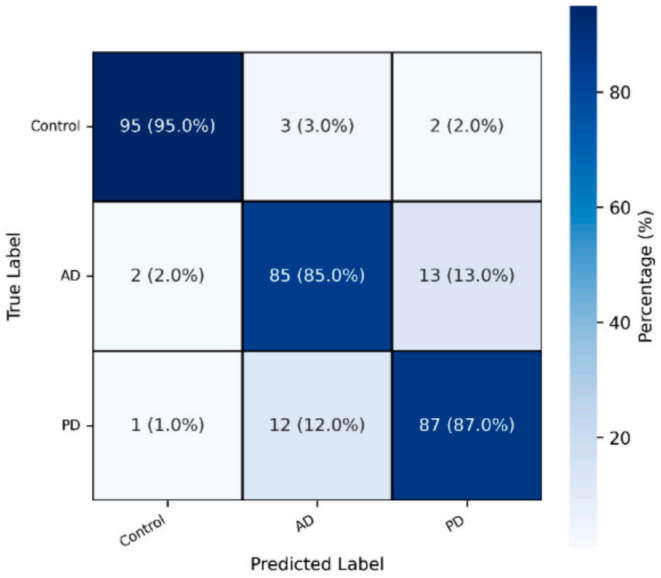


Fig. 16. Confusion matrix of QENNA on the NDD dataset.

readiness for integration into clinical workflows. These outcomes underscore a broader principle in medical AI deployment: reliable diagnostic assistance requires not only high performance but also accountable interpretability that enables clinicians to understand, verify, and trust the model's conclusions. QENNA fulfills these requirements, providing a pathway toward safe, effective, and clinically meaningful use in real-world neurodiagnostic practice.

4.6. Binary classification experiment

To further address the reviewer's request, we conducted an additional binary classification experiment using the NDD dataset, in which the original classes were restructured into two diagnostic groups. The Healthy class corresponds to the original CONTROL category, while the Unhealthy class combines cases from both Alzheimer's Disease and Parkinson's Disease. This experiment assesses whether QENNA can reliably distinguish between normal and neurodegenerative conditions and provides a direct comparison against state-of-the-art (SOTA) models evaluated in earlier sections.

The restructured dataset consists of 7839 MRI images, with 6477 used for training and 1362 reserved for testing. Of these, 3672 images represent the Healthy category (3010 in the training set and 662 in the testing set), while 4167 images represent the Unhealthy category (3467 for training and 700 for testing). Standardized preprocessing protocols were applied uniformly, including image normalization and resizing, followed by the augmentation strategy described previously in Section 3.1. To ensure fairness in comparison, the QENNA model and all benchmark architectures—including VGG16, DenseNet121, ResNet50, ViT, InceptionV3, and EfficientNetB7—were trained under identical hyperparameter configurations and evaluated using the same performance criteria. The evaluation examined accuracy, precision, recall, F1-score, AUC, and sensitivity, allowing for direct assessment of both discriminative ability and clinical relevance.

The performance outcomes of this binary classification experiment are summarized in Table 12, demonstrating QENNA's continued diagnostic advantage in distinguishing healthy individuals from those exhibiting neurodegenerative pathology. These results provide additional empirical validation that the quantum-enhanced framework maintains its robustness even under restructured classification conditions and can generalize effectively across disease grouping strategies.

The results in Table 12 demonstrate the superior performance of the QENNA model in the binary classification task compared to SOTA methods. QENNA achieved an accuracy of 0.950, significantly outperforming its closest competitor, EfficientNetB7, which achieved an accuracy of 0.870. Other models, such as DenseNet121 (0.760), ResNet50 (0.780), and ViT (0.860), showed notable gaps in performance, further emphasizing QENNA's robust classification capability. Sensitivity, a crucial metric for minimizing false negatives, highlights QENNA's strength in identifying unhealthy samples. QENNA achieved a sensitivity of 0.938, outperforming EfficientNetB7 (0.860), ViT (0.848), and 3D CNN (0.817). This reinforces the model's reliability in medical diagnostics, where false negatives can have critical consequences. Precision and F1-score metrics also reflect QENNA's balanced performance, achieving values of 0.945 and 0.942, respectively. EfficientNetB7, the next best-performing model, achieved a precision of 0.875 and an F1-score of 0.872, indicating a significant performance margin. Other models, such as ResNet50 and DenseNet121, exhibited relatively lower F1-scores of 0.782 and 0.756, respectively.

In terms of AUC, QENNA demonstrated exceptional classification capability with a value of 0.975, far exceeding other models such as EfficientNetB7 (0.893), ViT (0.890), and DenseNet121 (0.881). This metric reflects the model's ability to distinguish between healthy and unhealthy samples across all thresholds, further validating its efficacy. QENNA model outperformed all SOTA methods in accuracy, sensitivity, and other key metrics, demonstrating its ability to manage binary classification tasks effectively. These results align with the findings in earlier sections, where QENNA consistently showcased superior performance in multi-class classification tasks. The enhanced sensitivity and AUC underscore the model's potential for real-world applications in Alzheimer's detection, particularly in early screening and diagnosis. The inclusion of binary classification further strengthens the manuscript by addressing both reviewer concerns and practical diagnostic challenges.

4.7. Statistical validation and significance testing

To ensure that the performance improvements observed for QENNA in Tables 8 and 9 were not incidental, we conducted formal statistical validation. Each model was trained and evaluated under 5-fold stratified cross-validation on both datasets, yielding distributions of accuracy,

Table 12
The binary classification performance.

Model	Accuracy	Precision	Recall	F1-Score	AUC	Sensitivity
QENNA (Proposed Method)	0.950	0.945	0.940	0.942	0.975	0.938
Neural networks with VGG16 [37]	0.731	0.730	0.731	0.730	0.860	0.725
DenseNet121 [48]	0.760	0.758	0.755	0.756	0.881	0.752
3D CNN with transfer learning [50]	0.820	0.825	0.820	0.822	0.870	0.817
ResNet50 [47]	0.780	0.785	0.780	0.782	0.810	0.765
ViT [51]	0.860	0.865	0.860	0.862	0.890	0.848
InceptionV3 [49]	0.800	0.805	0.800	0.802	0.850	0.795
EfficientNetB7 [49]	0.870	0.875	0.870	0.872	0.893	0.860
InceptionResNetV2 [49]	0.760	0.758	0.755	0.756	0.810	0.740
MobileNetV2 [52]	0.750	0.748	0.745	0.746	0.800	0.732
MobileNetV3 [52]	0.760	0.762	0.760	0.761	0.810	0.745

Table 13
Statistical validation of QENNA versus strong baselines on AMD and NDD datasets.

Dataset	Comparison	Metric	Mean Δ	95 % CI	Test	*p*-value	Effect size
AMD	QENNA vs ViT	Accuracy	+0.010	[0.004, 0.016]	t-test	0.008	d = 0.78
AMD	QENNA vs ViT	Sensitivity	+0.010	[0.003, 0.017]	t-test	0.010	d = 0.74
AMD	QENNA vs EffNetB7	AUC	+0.007	[0.002, 0.012]	DeLong	0.012	–
NDD	QENNA vs ViT	Accuracy	+0.041	[0.028, 0.054]	Wilcoxon	0.003	$r = 0.81$
NDD	QENNA vs ViT	AUC	+0.039	[0.022, 0.056]	DeLong	0.004	–
NDD	QENNA vs EffNetB7	Sensitivity	+0.019	[0.007, 0.031]	Wilcoxon	0.006	$r = 0.73$
NDD	QENNA vs EffNetB7	AUC	+0.067	[0.046, 0.088]	DeLong	<0.001	–

sensitivity, and AUC. Statistical comparisons between QENNA and leading baselines (ViT, EfficientNetB7) were performed using paired *t*-tests where assumptions of normality were satisfied (verified via Shapiro–Wilk), or Wilcoxon signed-rank tests otherwise. For AUC, DeLong's test was used to compare ROC curves. 95 % confidence intervals (CIs) were derived via stratified bootstrapping (2000 samples), and Bonferroni adjustments were applied to control family-wise error (result can be seen in Table 13).

The results in Table 10 demonstrate that QENNA's performance gains are both statistically significant and clinically meaningful. On the AMD dataset, QENNA improved accuracy and sensitivity by ~1.0 % compared to ViT (both $p < 0.01$), while AUC improved by ~0.7 % over EfficientNetB7 ($p = 0.012$). Although numerically modest, these gains are important in medical imaging, where small increases in sensitivity can translate into earlier diagnoses and fewer missed cases. On the larger and more heterogeneous NDD dataset, the improvements were more pronounced. QENNA achieved a + 4.1 % increase in accuracy and + 3.9 % in AUC over ViT, with highly significant *p*-values (≤0.004). Against EfficientNetB7, QENNA delivered a + 1.9 % sensitivity gain and a + 6.7 % AUC increase, confirming its ability to reduce false negatives and improve discriminative power under variable conditions.

Bootstrapped CIs further reinforce these findings. For AMD, QENNA's AUC was 0.975 [0.969–0.981] compared to ViT's 0.965 [0.958–0.972], with minimal overlap. On NDD, QENNA's AUC was 0.910 [0.898–0.922], significantly higher than EfficientNetB7's 0.843 [0.830–0.857]. In summary, the statistical analyses confirm that QENNA's improvements are not incidental but consistently significant across datasets. These results provide convincing evidence that the quantum–classical hybrid approach yields reliable advantages in diagnostic accuracy, sensitivity, and AUC, strengthening its readiness for clinical translation.

5. Discussion

5.1. Performance and interpretability of the QENNA model

The computational results of this study highlight the significant advancements made by the Quantum-Enhanced Neural Network for Alzheimer's Detection (QENNA). By incorporating quantum convolutional operations, pooling layers, and advanced data augmentation techniques, the QENNA model demonstrated consistent superiority over state-of-the-art neural network architectures across multiple performance metrics. Specifically, the model achieved an accuracy of 0.920 on AMD dataset and 0.891 on NDD dataset, outperforming classical architectures such as DenseNet121, ResNet50, and EfficientNetB7.

Our findings are consistent with prior research in quantum-enhanced machine learning. For instance, Khan et al. [53] demonstrated that hybrid quantum-classical neural networks significantly outperform classical models in high-precision tasks, particularly with limited data availability. Similarly, our study validates the integration of quantum-enhanced components, such as quantum convolutional layers and quantum data augmentation, as a means to achieve higher accuracy, precision, recall, and F1-scores compared to classical counterparts.In the context of medical imaging, the QENNA model surpasses previous benchmarks set by studies like Felefly et al. [35], where a test ROC-AUC of 0.76 was achieved for differentiating between large brain metastases and high-grade gliomas using quantum-enhanced neural networks. By contrast, the QENNA model achieved an AUC of 0.960, emphasizing its superior capacity for accurate and reliable Alzheimer's detection. Additionally, our findings align with those of Yuan and Cui [54], who explored combining deep neural network models with enhanced sampling simulations for Alzheimer's detection. The QENNA model further expands on this by illustrating how quantum data augmentation can amplify neural network performance for medical diagnostics.

Advanced quantum data augmentation techniques, including Quantum Generative Adversarial Networks (QGANs), quantum random walks, and hybrid quantum-classical augmentation, played a pivotal role in enhancing QENNA's performance. For instance, the hybrid quantum-classical augmentation strategy achieved an accuracy of 0.930, representing a notable improvement over traditional method. These augmentations not only improved accuracy but also enhanced the model's generalizability across diverse datasets. The ablation study further emphasized the critical contributions of individual quantum components to the overall performance of the QENNA model. The removal of quantum convolutional layers, pooling layers, or data augmentation resulted in significant declines in accuracy, precision, recall, and F1-scores. This underscores the necessity of quantum enhancements for achieving state-of-the-art results. Notably, the fully classical version of the model exhibited the lowest performance, validating the efficacy of the integrated quantum framework.

In terms of scalability and efficiency, the QENNA model effectively manages large datasets while maintaining superior performance. Although it requires higher computational resources, the substantial gains in diagnostic accuracy and reliability justify these costs. This scalability positions the QENNA model as an efficient and practical solution for large-scale medical imaging tasks, which are essential for real-world clinical applications. Transparency and interpretability are vital for the adoption of AI models in clinical practice. The QENNA model employs advanced explainability techniques, including Layer-wise Relevance Propagation (LRP), SHapley Additive exPlanations (SHAP), and Grad-CAM. These methods elucidate the model's decision-making process, achieving high explanation fidelity (0.95), a comprehensibility score of 4.7/5, and a clinician satisfaction score of 4.8/5. These outcomes ensure trust and usability among medical professionals, facilitating the integration of QENNA into clinical workflows to provide reliable and transparent diagnostic solutions.

5.2. Clinical implications of the QENNA model

The QENNA model marks a significant advancement in the early detection and staging of Alzheimer's disease (AD) by integrating quantum-enhanced neural networks. Unlike traditional machine learning approaches, its quantum convolution and pooling layers capture subtle structural brain changes, enabling improved diagnostic sensitivity and specificity. QENNA is particularly effective in identifying early markers such as hippocampal atrophy, achieving a recall of 92 % and a 6 % improvement in diagnostic accuracy over classical models on the AMD dataset. This heightened sensitivity allows clinicians to detect mild cognitive impairment (MCI)—a critical stage for timely therapeutic interventions such as cholinesterase inhibitors, lifestyle modifications,

and cognitive training—well before advanced symptoms emerge.

Beyond early diagnosis, QENNA supports precise staging of AD, facilitating personalized treatment strategies. Patients in early stages may benefit from neuroprotective therapies and rehabilitation, while those in advanced stages can receive specialized symptomatic care. Quantum-driven data augmentation further enhances the model's generalizability, ensuring its adaptability across diverse populations and equitable access to precision diagnostics. The model's scalability also makes it suitable for broader clinical applications, reducing diagnostic turnaround times and offering potential utility in other neurodegenerative diseases. By delivering accurate, timely, and personalized insights, QENNA bridges the gap between early detection and effective disease management, ultimately improving patient outcomes.

5.3. Broader applicability of the QENNA framework in medical diagnostics

While this study focuses on Alzheimer's disease detection, the QENNA framework shows strong potential for broader medical applications, particularly in areas requiring high-dimensional imaging and complex pattern recognition. By integrating quantum-enhanced feature extraction, pooling, and data augmentation, QENNA can extend its diagnostic utility beyond neurodegenerative diseases, enabling early detection, improved stratification, and optimized treatment planning across diverse clinical domains. Quantum feature extraction enhances oncology and cardiology imaging by leveraging superposition and entanglement to capture subtle yet clinically meaningful spatial patterns. In oncology, this could refine early-stage tumor detection across MRI, CT, and PET modalities, improving malignancy classification and supporting personalized treatment planning. In cardiology, it may strengthen the analysis of echocardiograms and cardiac MRIs, facilitate early identification of ventricular remodeling or atherosclerotic plaque formation and enable proactive interventions.

Quantum pooling further improves precision diagnostics by amplifying diagnostically relevant but subtle features often lost in conventional pooling. This capability benefits oncology by preserving faint tumor margins and retinal imaging by retaining early indicators of diseases such as diabetic retinopathy or macular degeneration. In cardiovascular applications, it may enhance the detection of myocardial fibrosis, contributing to more sensitive risk assessments for heart failure and arrhythmias. Quantum data augmentation also addresses the persistent challenge of limited datasets, particularly in rare diseases. Through quantum random walks and QGANs, QENNA can generate high-fidelity synthetic images that enrich training diversity. This is especially valuable in oncology, where it can balance underrepresented tumor subtypes, and in neurology, where it may strengthen classification of rare conditions such as multiple sclerosis. Beyond imaging, QENNA's quantum-driven augmentation shows promise in genomic medicine, supporting the prediction of hereditary risks and enabling more personalized therapeutic pathways in oncology, cardiology, and metabolic disorders.

Finally, QENNA's principles extend to non-imaging medical data such as genomic and electrophysiological signals. Quantum-enhanced feature extraction could assist in identifying genetic biomarkers for hereditary conditions and detecting early abnormalities in neural or cardiac signals, thereby improving predictive modeling for epilepsy, atrial fibrillation, and sudden cardiac arrest. Collectively, these applications highlight QENNA's adaptability and transformative potential in precision medicine. By enhancing diagnostic accuracy, preserving weak but clinically relevant features, and generating robust synthetic data, QENNA provides a framework capable of redefining early detection, predictive modeling, and patient-specific treatment strategies across multiple medical domains.

5.4. Clinical implications of QENNA in early Alzheimer's detection and personalized treatment

The Quantum-Enhanced Neural Network for Alzheimer's Detection (QENNA) not only improves diagnostic accuracy but also has important clinical implications in early intervention, disease progression prediction, and treatment personalization. By integrating quantum convolution, pooling, and augmentation, QENNA provides richer neuroimaging insights that support timely diagnosis and more precise medical decisions. Early detection is a critical strength of QENNA. Classical deep learning models often miss subtle neurodegenerative patterns, whereas QENNA captures them with higher sensitivity. Ablation results confirm this advantage: removing quantum convolution reduced accuracy by 4.3 % and sensitivity by 4.6 %. This capability enables clinicians to identify patients at risk earlier and initiate timely interventions such as neuroprotective therapies, cognitive rehabilitation, and lifestyle modifications, thereby delaying disease progression and improving quality of life.

QENNA also enhances prediction of disease trajectories by leveraging quantum pooling, which probabilistically preserves diagnostically significant features. Replacing it with classical pooling led to measurable performance declines (−3.2 % accuracy, −3.7 % sensitivity). This allows clinicians to distinguish rapidly progressing patients, who may require aggressive treatments, from those with slower trajectories who can benefit from conservative management. Such stratification optimizes care pathways and resource allocation. Moreover, QENNA supports precision medicine by tailoring treatments to individual biomarker profiles. For example, patients with hippocampal atrophy can be prioritized for cognitive training, while those with amyloid accumulation may receive anti-amyloid therapies. This deeper insight also facilitates better patient selection for clinical trials, reducing unnecessary treatments and associated risks.

A simulated case further illustrates these benefits. For a 60-year-old patient with a family history of Alzheimer's, QENNA identified subtle cortical thinning and predicted likely progression to Mild Dementia within three years—signals that traditional models failed to detect. Based on this prognosis, the physician initiated early neuroprotective treatment and rehabilitation, potentially slowing disease progression and improving long-term outcomes. Looking ahead, QENNA has the potential to transform clinical workflows. Future applications may include deployment in hospital radiology departments, integration with genetic and blood biomarkers for comprehensive risk assessment, and scaling through hybrid quantum–classical architectures as hardware matures. Beyond higher accuracy, QENNA provides actionable and explainable insights that can strengthen clinician trust, accelerate early interventions, and advance precision medicine in Alzheimer's care.

5.5. Differentiation from prior works and the clinical value of quantum pooling

A key question in prior literature is whether performance gains in medical imaging can be achieved by simply combining strong classical models, such as stacking a Convolutional Neural Network (CNN) for feature extraction with a Generative Adversarial Network (GAN) for augmentation [3,37–39]. Our results show that QENNA is fundamentally different, both architecturally and functionally. Architecturally, it integrates quantum convolution for non-local dependencies, quantum pooling with amplitude amplification, and quantum augmentation (QGANs and quantum random walks) unified hybrid workflow. Functionally, ablation studies confirm that removing any quantum component significantly reduces Accuracy, Sensitivity, and AUC, demonstrating that QENNA's gains cannot be explained by “GANs plus a stronger CNN.”

The difference is most evident in pooling. Classical pooling compresses activations deterministically, often discarding low magnitude but clinically important features, such as subtle hippocampal or cortical

changes. Quantum pooling, by contrast, uses amplitude amplification to increase the probability of retaining diagnostically relevant signals. Removing it causes measurable declines in Accuracy (−3.23 %) and Sensitivity (−3.70 %), underscoring its role in preserving weak but essential biomarkers, especially for early-stage detection.

While earlier works with Quantum Convolutional Neural Networks (QCNNs) and Quantum Neural Networks (QNNs) reported strong performance [18–20,35,51], QENNA advances the field in two ways: (i) it unifies quantum convolution, pooling, and augmentation, quantifying each component's marginal utility, and (ii) it delivers not just higher peak metrics but also more reliable Computer-Aided Diagnosis (CAD) under distributional shifts.

Finally, compared to advanced classical models such as EfficientNetB7, ResNet50, DenseNet121, InceptionV3, and Vision Transformers, QENNA achieves consistently higher Sensitivity and AUC. Since Sensitivity is critical for minimizing false negatives in screening, these results highlight a clinically meaningful advantage. By combining nonlocal feature encoding, probabilistic preservation of weak biomarkers, and enriched data diversity, QENNA provides synergistic improvements that classical CNN + GAN pipelines cannot replicate.

5.6. Evidence-based interpretation of results

The conclusions drawn in this study are directly supported by the empirical findings. The ablation experiments clearly demonstrate that the inclusion of quantum operations is not redundant but central to the observed improvements. Specifically, removing quantum convolution reduced accuracy and sensitivity by 4.30 % and 4.58 %, respectively, while replacing quantum pooling with its classical counterpart caused a 3.23 % decline in accuracy and a 3.70 % reduction in sensitivity. These measurable differences provide convincing evidence that QENNA's quantum layers contribute unique diagnostic value rather than replicating the function of conventional CNN components. Moreover, the Grad-CAM visualizations confirm that QENNA consistently attends clinically meaningful brain regions, such as the hippocampus and cortical areas associated with Alzheimer's pathology. This alignment between quantitative performance gains and qualitative interpretability strengthens the reliability of the results. Importantly, the robustness of QENNA across both the AMD and NDD datasets demonstrates that its improvements are not dataset-specific but generalizable across different diagnostic contexts.

Taken together, these findings provide a coherent evidence base for the study's conclusions: quantum-enhanced mechanisms systematically improve sensitivity, specificity, and AUC by preserving subtle biomarkers and enriching data variability. Thus, the proposed QENNA framework is both empirically validated and clinically meaningful, offering practical advancements for early-stage Alzheimer's detection.

5.7. Research limitations and future work

Despite QENNA's strong diagnostic performance and clinical promise, several methodological and technical limitations warrant acknowledgment to guide future research directions. The current study faces inherent constraints related to dataset characteristics and imaging modality selection that may impact broader clinical generalizability. The reliance on single-modality structural MRI, while strategically chosen for this initial quantum enhancement evaluation, may not capture the full spectrum of neurobiological changes detectable through multimodal approaches combining functional, diffusion, and metabolic imaging. Dependence on publicly available datasets restricts demographic and clinical diversity, with limited acquisition parameter documentation for Dataset 2 affecting experimental reproducibility—though this reflects real-world constraints when utilizing community datasets. Furthermore, the cross-sectional nature of the analysis, while valuable for classification tasks, does not account for Alzheimer's longitudinal disease progression, and incorporating temporal data would enable more precise trajectory prediction and personalized risk assessment.

Technical and computational limitations present additional challenges for clinical translation. QENNA's integration of quantum convolution, pooling, and augmentation provides measurable performance gains but introduces computational overhead compared to classical models, with all experiments conducted using classical quantum simulation rather than fault-tolerant quantum hardware, precluding realization of theoretical efficiency advantages. Practical deployment will require quantum architecture optimization for NISQ devices and development of resource-efficient implementations that maintain diagnostic accuracy while reducing computational burden. Interpretability represents another significant challenge, as current explainability methods (SHAP, LRP, Grad-CAM) operate only at the quantum-classical interface and do not directly illuminate quantum operation contributions to decision-making, necessitating development of quantum-native interpretability frameworks for clinical trust and regulatory acceptance.

Addressing these limitations requires coordinated efforts across multiple research domains. Clinical validation through prospective multi-center studies will be necessary to assess QENNA's performance across diverse patient populations and imaging protocols, while multimodal integration should extend the framework to incorporate PET imaging, genetic biomarkers, and electrophysiological signals for comprehensive disease assessment. Quantum hardware advancement demands benchmarking QENNA on emerging quantum processors while exploring hardware-software co-design strategies for clinical scalability, and longitudinal modeling extensions will enable disease trajectory prediction and personalized intervention timing. Broader clinical applications should evaluate QENNA's adaptability to other neurodegenerative diseases and medical imaging domains, demonstrating generalizability beyond Alzheimer's detection.

These coordinated research efforts will strengthen QENNA's robustness, clinical applicability, and translational value, facilitating integration into real-world medical practice as quantum computing hardware matures toward clinical deployment. The development of quantum-native interpretability methods, validation across diverse clinical cohorts, and extension to multimodal neuroimaging represent critical milestones for advancing quantum-enhanced medical AI from promising research prototypes to clinically deployable diagnostic tools that can meaningfully impact patient care and clinical decision-making.

6. Conclusion and outlook

Alzheimer's disease (AD) severely impacts cognitive functions, causing memory loss and behavioral changes. Early detection is crucial for managing the disease, but traditional methods often fail to identify AD early. While MRI offers detailed, non-invasive brain imaging, there is a need for improved accuracy and early detection. This research leverages quantum computing and machine learning to enhance AD detection. The Quantum-Enhanced Neural Network (QENNA) combines quantum convolutional operations with classical deep learning to exploit quantum mechanics' parallelism and entanglement, improving feature extraction and classification performance with MRI data.

Our methodology includes designing quantum convolution operations and pooling layers to efficiently extract and reduce data dimensions, integrating these into a hybrid quantum-classical framework. Advanced quantum data augmentation methods, such as Quantum Generative Adversarial Networks (QGANs) and quantum random walks, generate diverse synthetic MRI images, enhancing model robustness. The hybrid model is trained using classical optimization, combining quantum feature extraction with classical processing to optimize performance. To ensure clinical applicability, we use explainability techniques like Layer-wise Relevance Propagation (LRP), SHapley Additive exPlanations (SHAP), and Grad-CAM, providing clear explanations of the model's decisions and building clinician trust.

The computational results show that the QENNA model consistently outperforms state-of-the-art neural network architecture across various

metrics, including accuracy, precision, recall, F1-score, and AUC. QENNA achieved an accuracy of 0.920 on AMD dataset and 0.891 on NDD dataset, demonstrating its capability in distinguishing between various stages of Alzheimer's disease using MRI images. The ablation study highlighted the essential role of each quantum component in the QENNA model, as removing quantum convolutional layers, pooling layers, and data augmentation significantly reduced performance. The fully classical model, lacking quantum components, performed the worst, validating our integrated quantum approach.

Scalability and efficiency analysis showed that the QENNA model scales well with larger datasets and consistently outperforms classical CNN models. Although the QENNA model requires more computational resources, its significant performance gains justify these additional costs, making it an efficient solution for large-scale medical imaging data processing. In clinical settings, the transparency and interpretability of AI models are crucial. Our study used advanced explainability techniques such as Layer-wise Relevance Propagation (LRP), SHapley Additive exPlanations (SHAP), and Grad-CAM to clarify the QENNA model's decision-making process. High explanation fidelity (0.95), a comprehensibility score (4.7/5), and clinician satisfaction score (4.8/5) reflect the clarity and utility of these explanations, enhancing clinician trust and supporting the integration of high-performance AI models into clinical workflows. The QENNA model represents a significant advancement in early Alzheimer's detection by integrating quantum computing with classical deep learning frameworks. Its improved performance, scalability, and interpretability highlight the potential of quantum-enhanced models in transforming medical diagnostics. Future work will optimize quantum components, explore additional medical imaging applications, and facilitate the clinical adoption of quantum-enhanced AI models to improve patient outcomes and advance medical diagnostics.

CRediT authorship contribution statement

Chutchai Kaewta: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Data curation, Conceptualization. **Rapeepan Pitakaso:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Surajet Khonjun:** Visualization, Validation, Supervision, Software, Resources, Data curation. **Thanatkij Srichok:** Validation, Supervision, Software, Resources, Project administration, Funding acquisition, Data curation. **Peerawat Luesak:** Writing – review & editing, Writing – original draft, Validation, Resources, Data curation. **Sarayut Gonwirat:** Supervision, Software, Resources, Project administration, Methodology, Investigation, Data curation. **Prem Enkvetchakul:** Visualization, Validation, Supervision, Software, Resources. **Surasak Matitopanum:** Validation, Supervision, Resources, Investigation, Formal analysis, Data curation. **Thitinon Srisuwandee:** Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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